

BFS Traversal

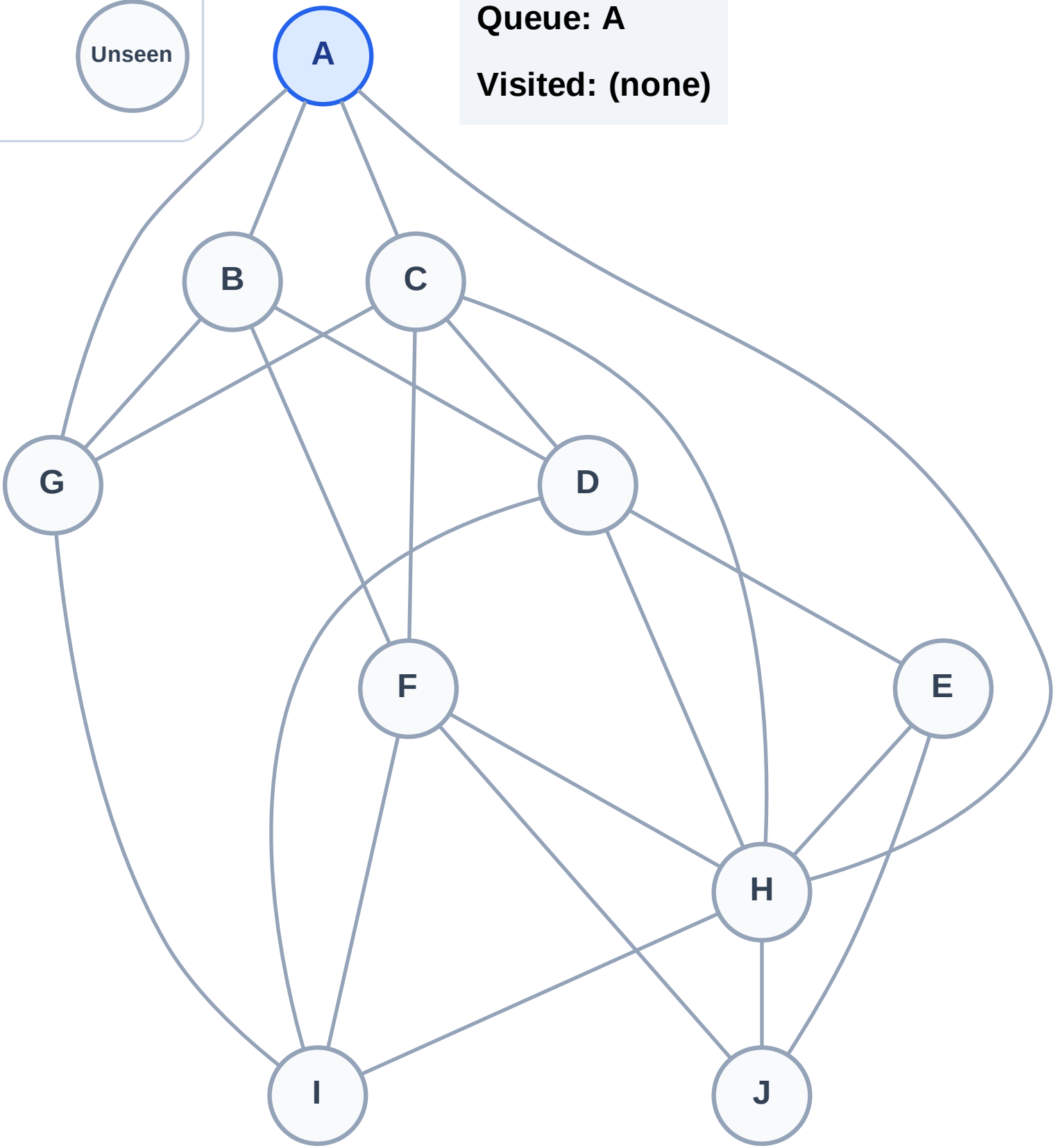
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

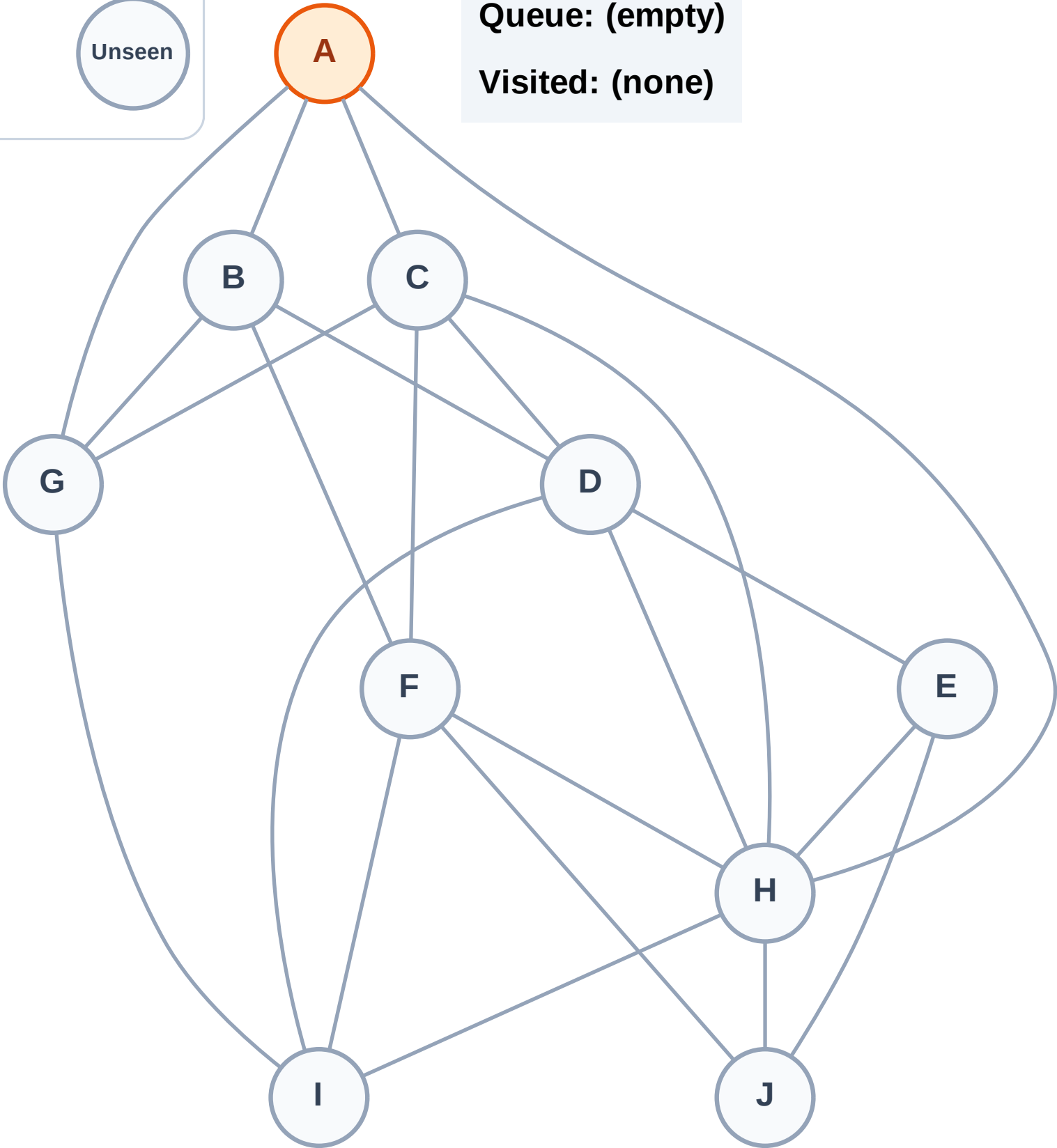
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover B, C, G, H from A

Legend

Complete

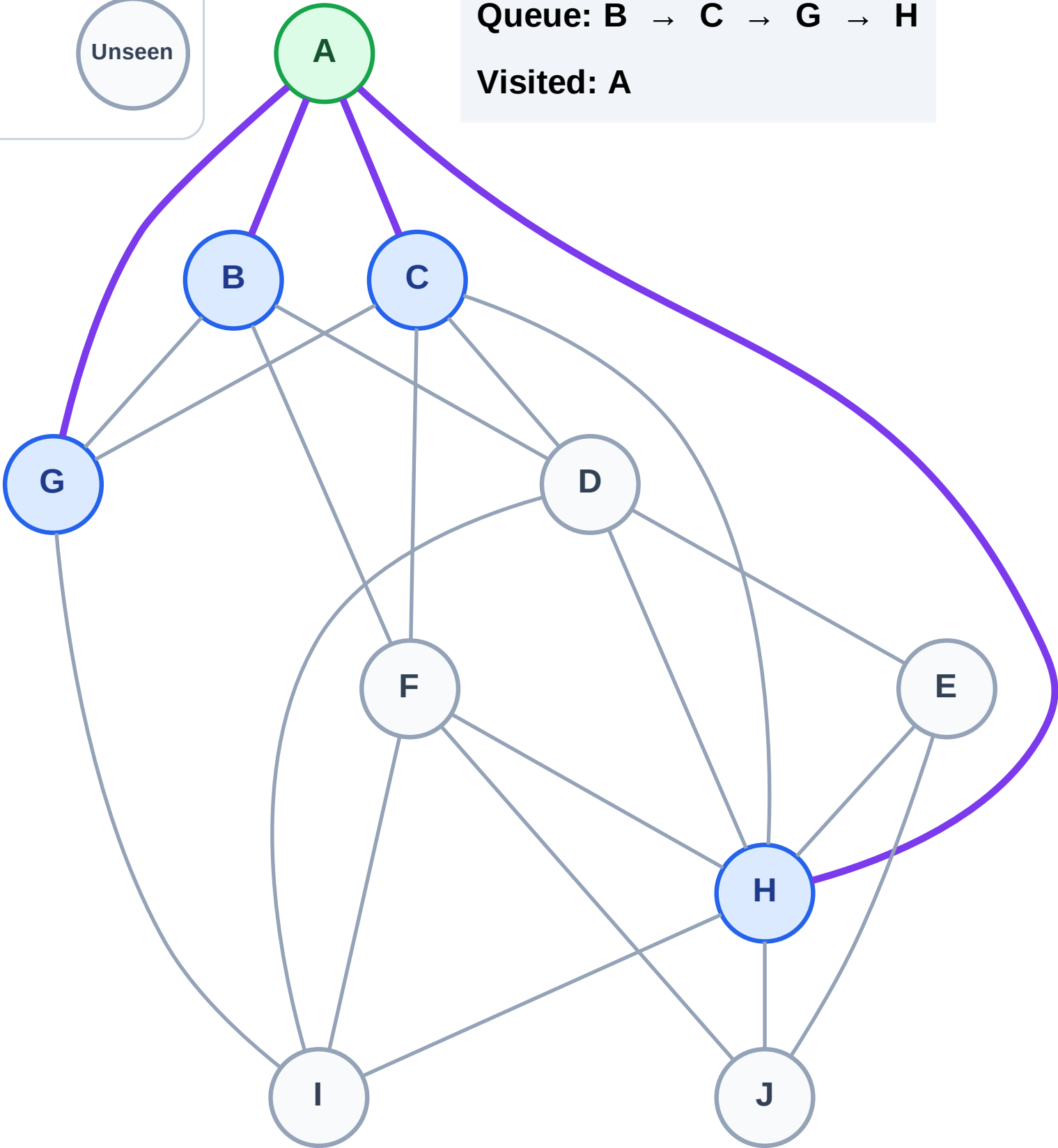
Processing

Discovered

Unseen

Queue: B → C → G → H

Visited: A



BFS Traversal

Step 4: Process node B

Legend

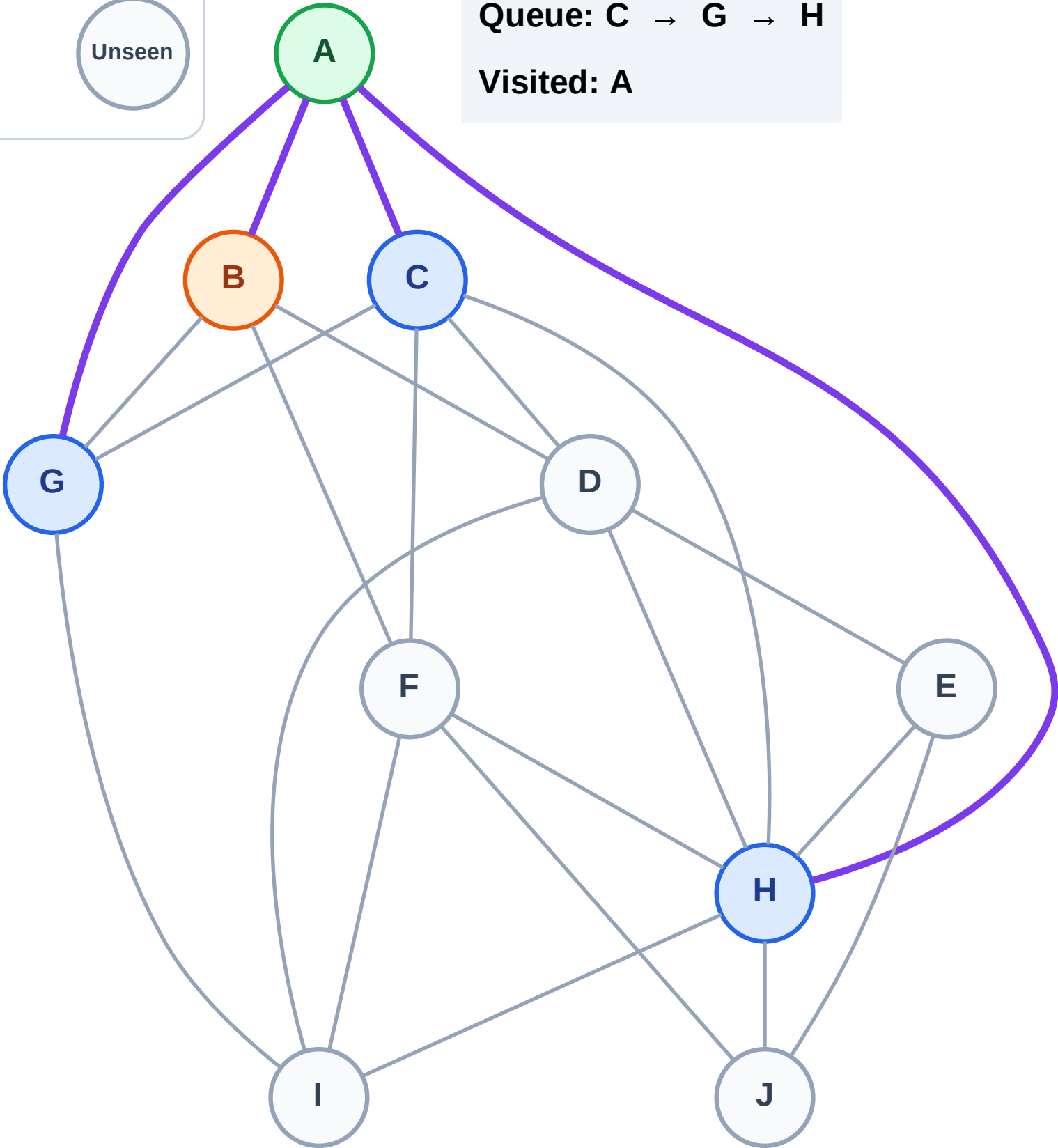
Complete

Processing

Discovered

Unseen

Queue: C → G → H
Visited: A



BFS Traversal

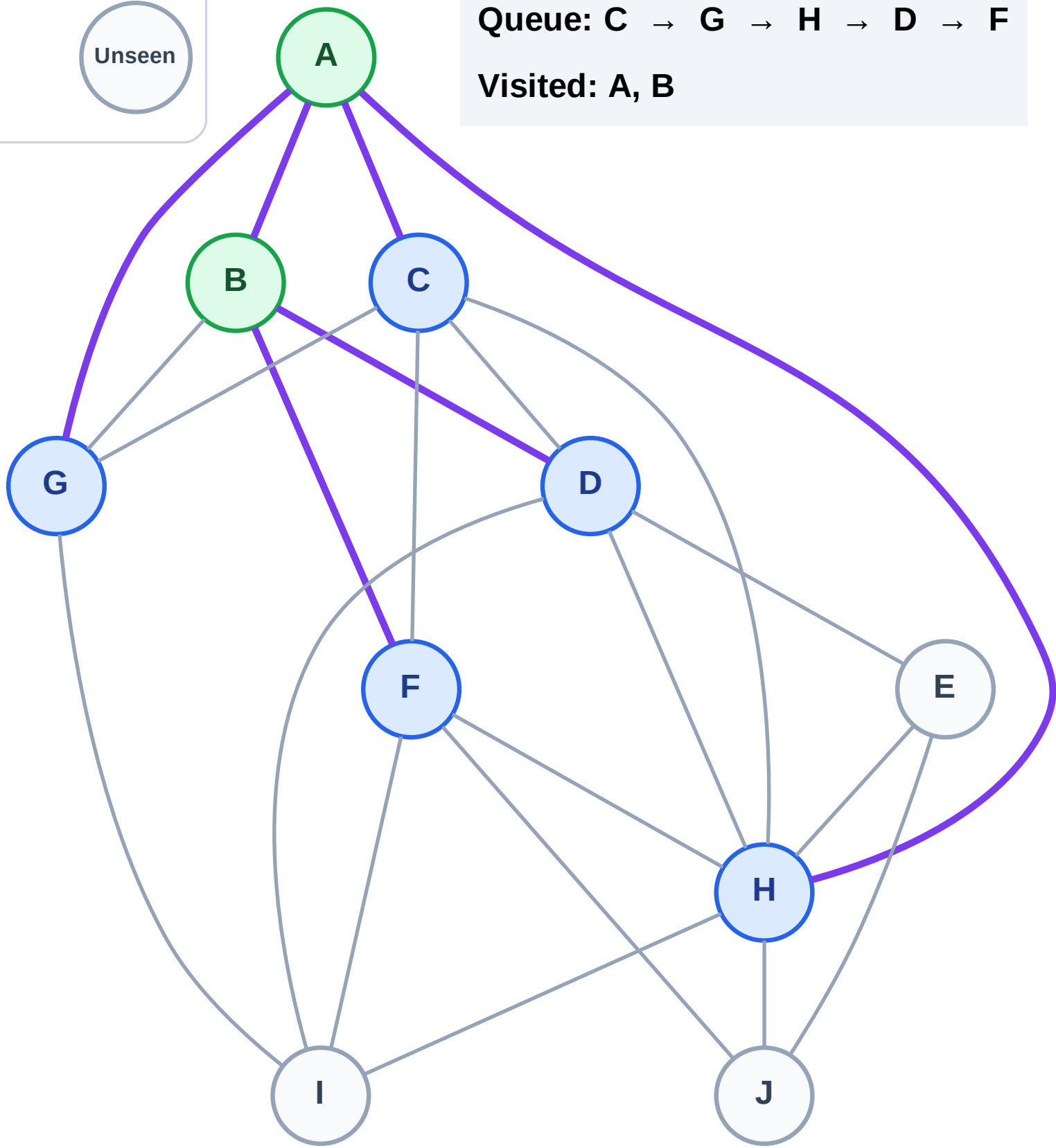
Step 5: Discover D, F from B

Legend



Queue: C → G → H → D → F

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

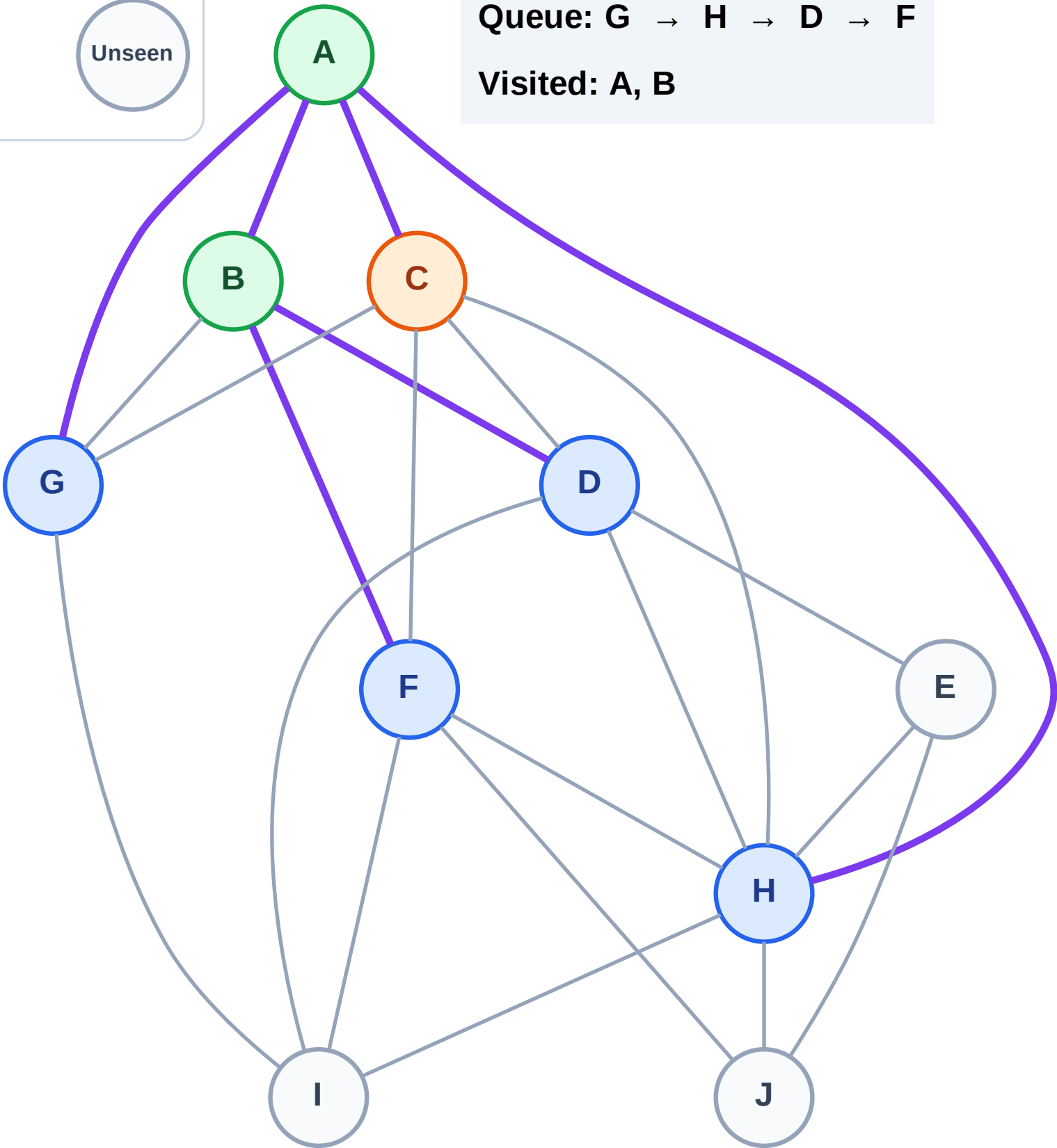
Processing

Discovered

Unseen

Queue: G → H → D → F

Visited: A, B



BFS Traversal

Step 7: No new neighbors from C

Legend

Complete

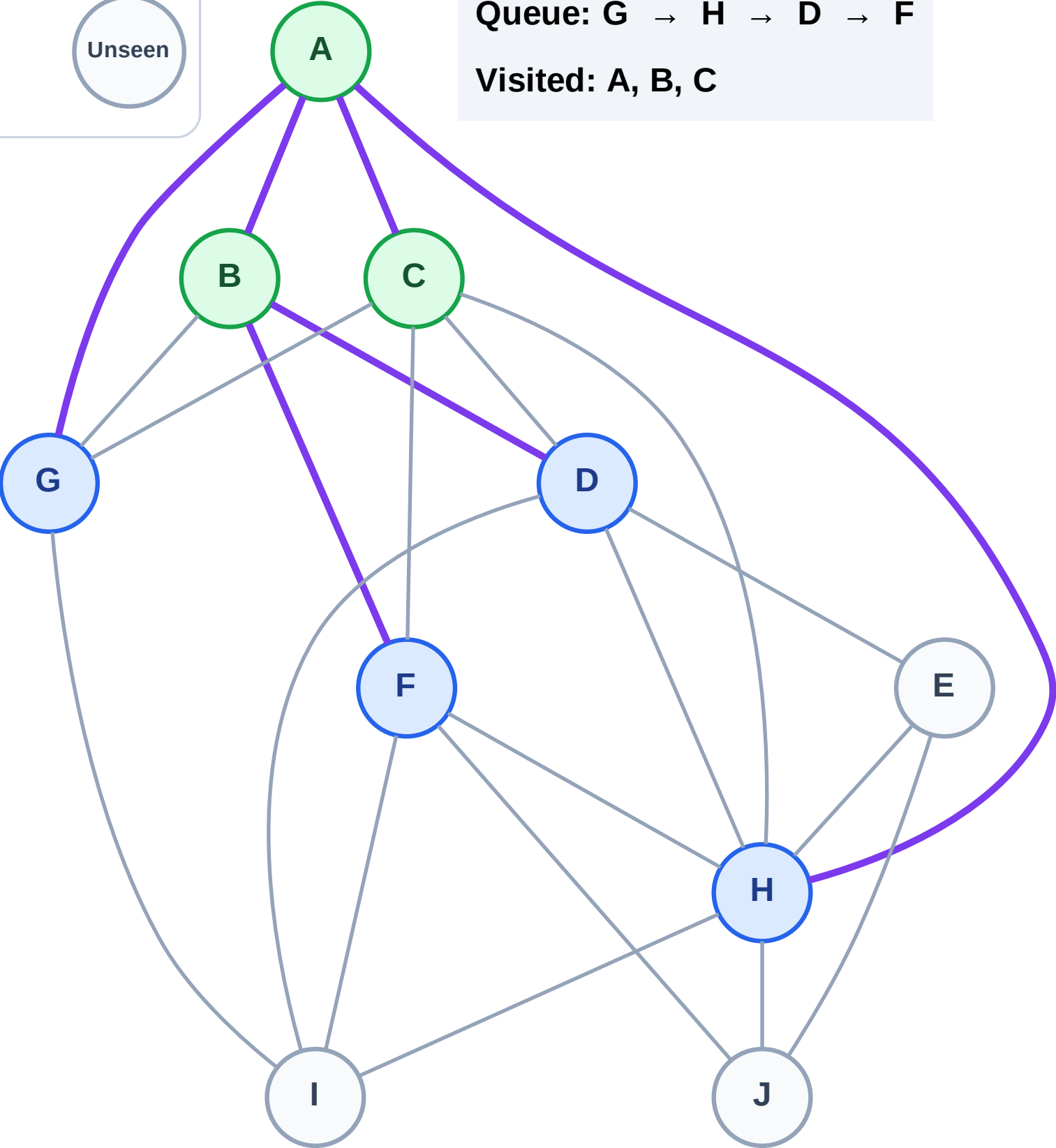
Processing

Discovered

Unseen

Queue: G → H → D → F

Visited: A, B, C



BFS Traversal

Step 8: Process node G

Legend

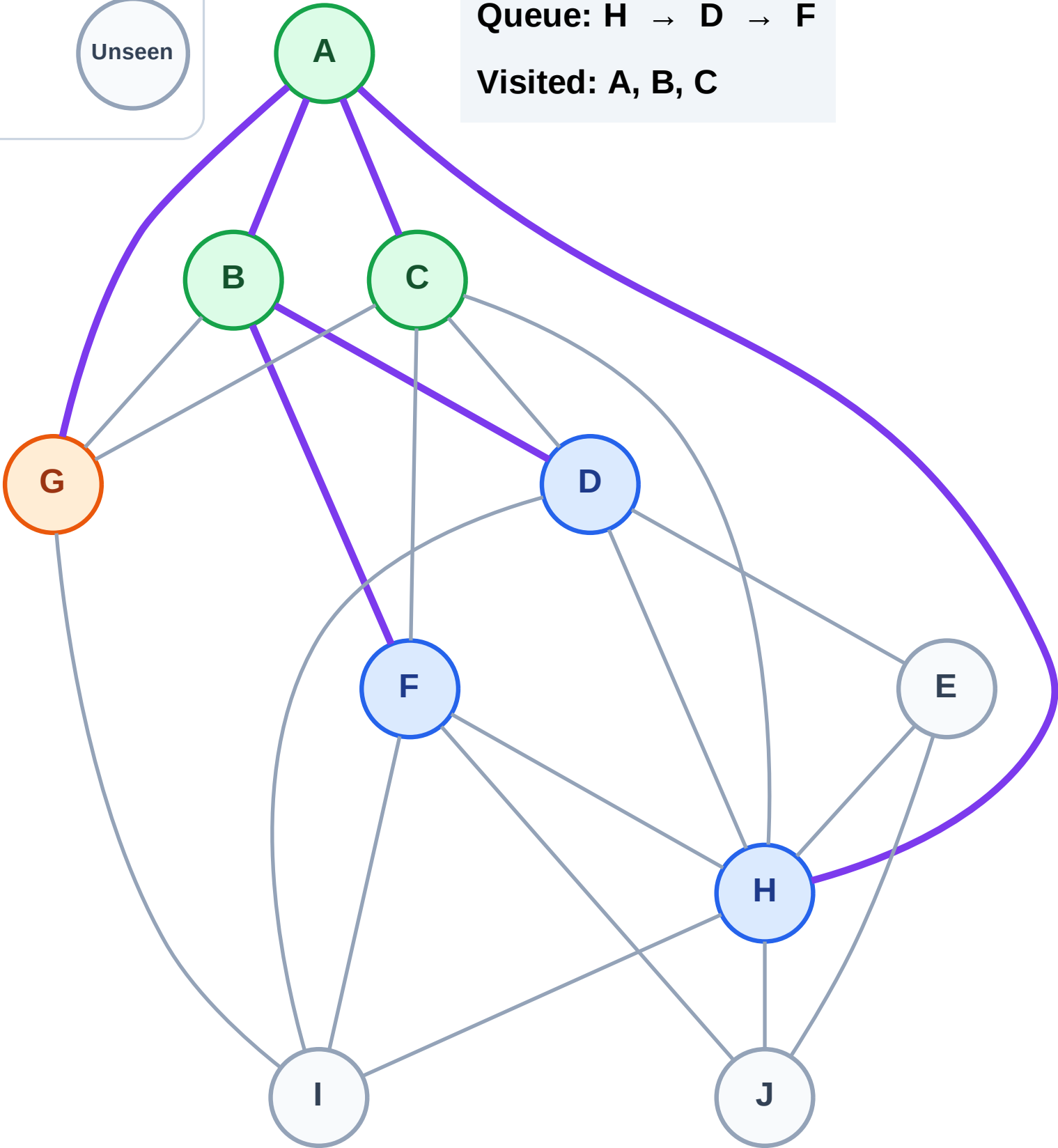
Complete

Processing

Discovered

Unseen

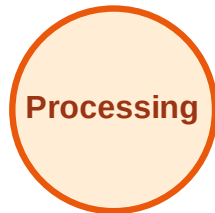
Queue: H → D → F
Visited: A, B, C



BFS Traversal

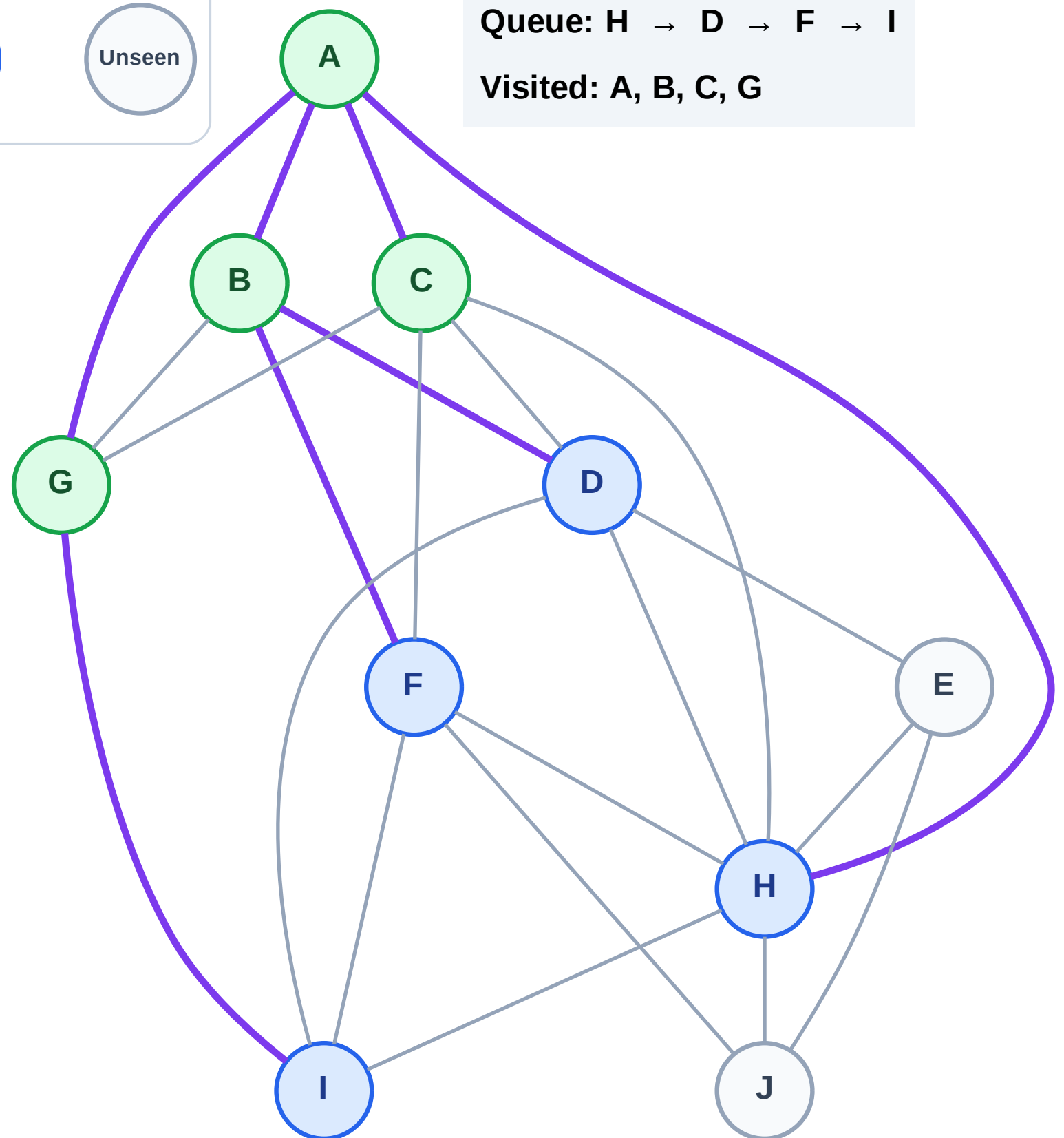
Step 9: Discover I from G

Legend



Queue: H → D → F → I

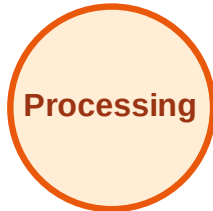
Visited: A, B, C, G



BFS Traversal

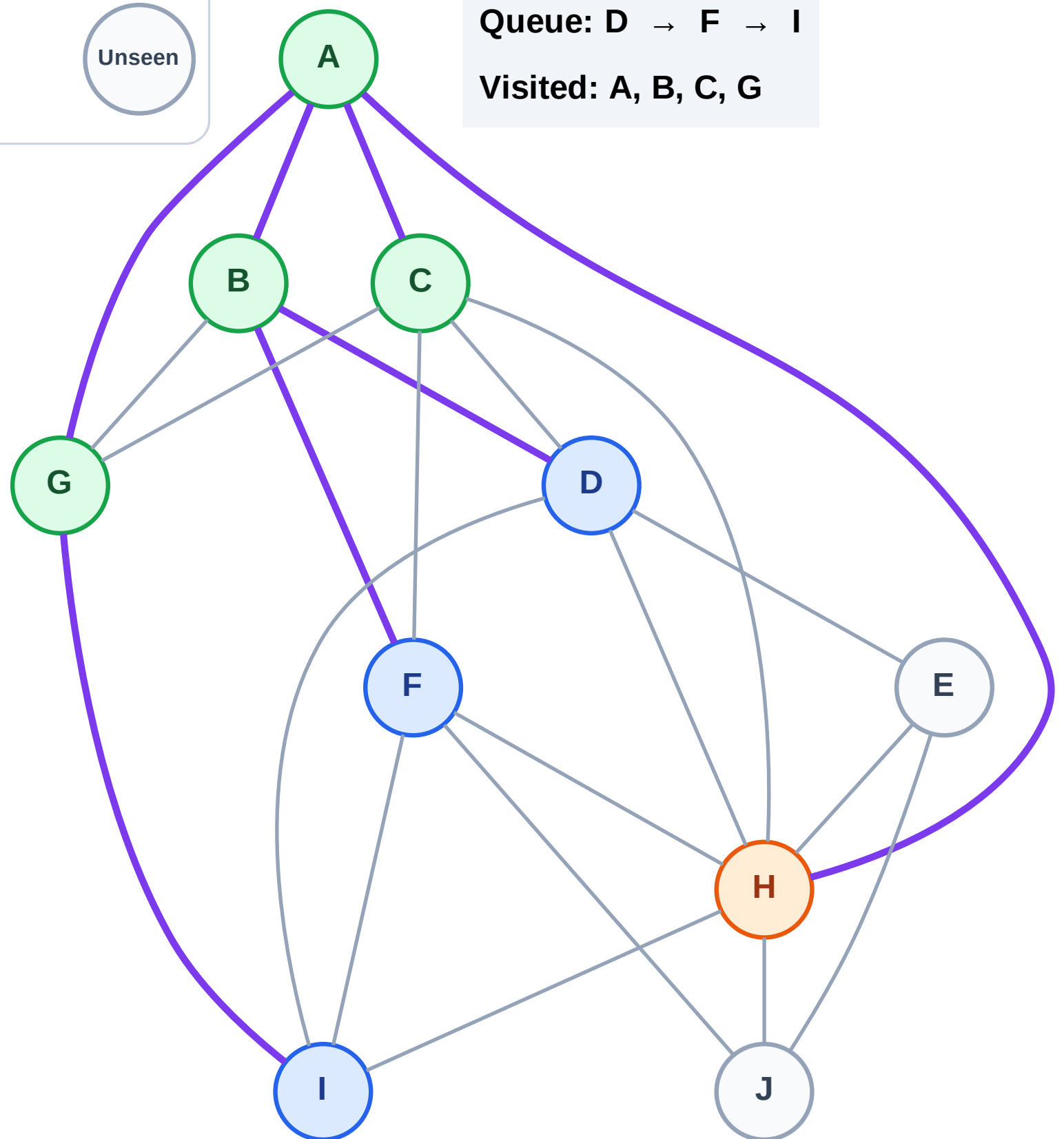
Step 10: Process node H

Legend



Queue: D → F → I

Visited: A, B, C, G



BFS Traversal

Step 11: Discover E, J from H

Legend

Complete

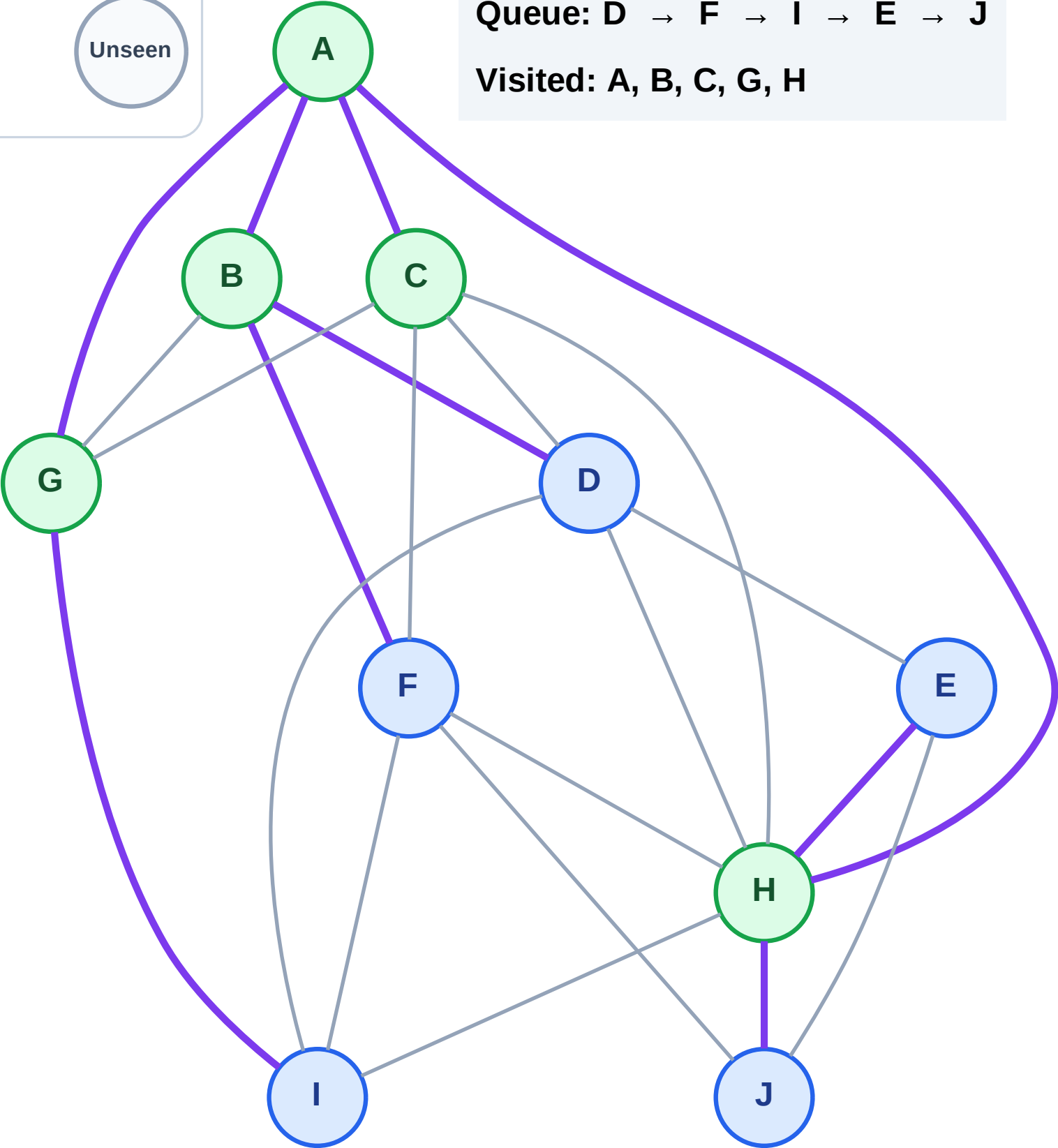
Processing

Discovered

Unseen

Queue: D → F → I → E → J

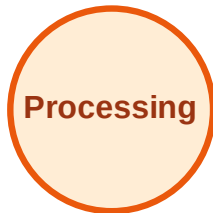
Visited: A, B, C, G, H



BFS Traversal

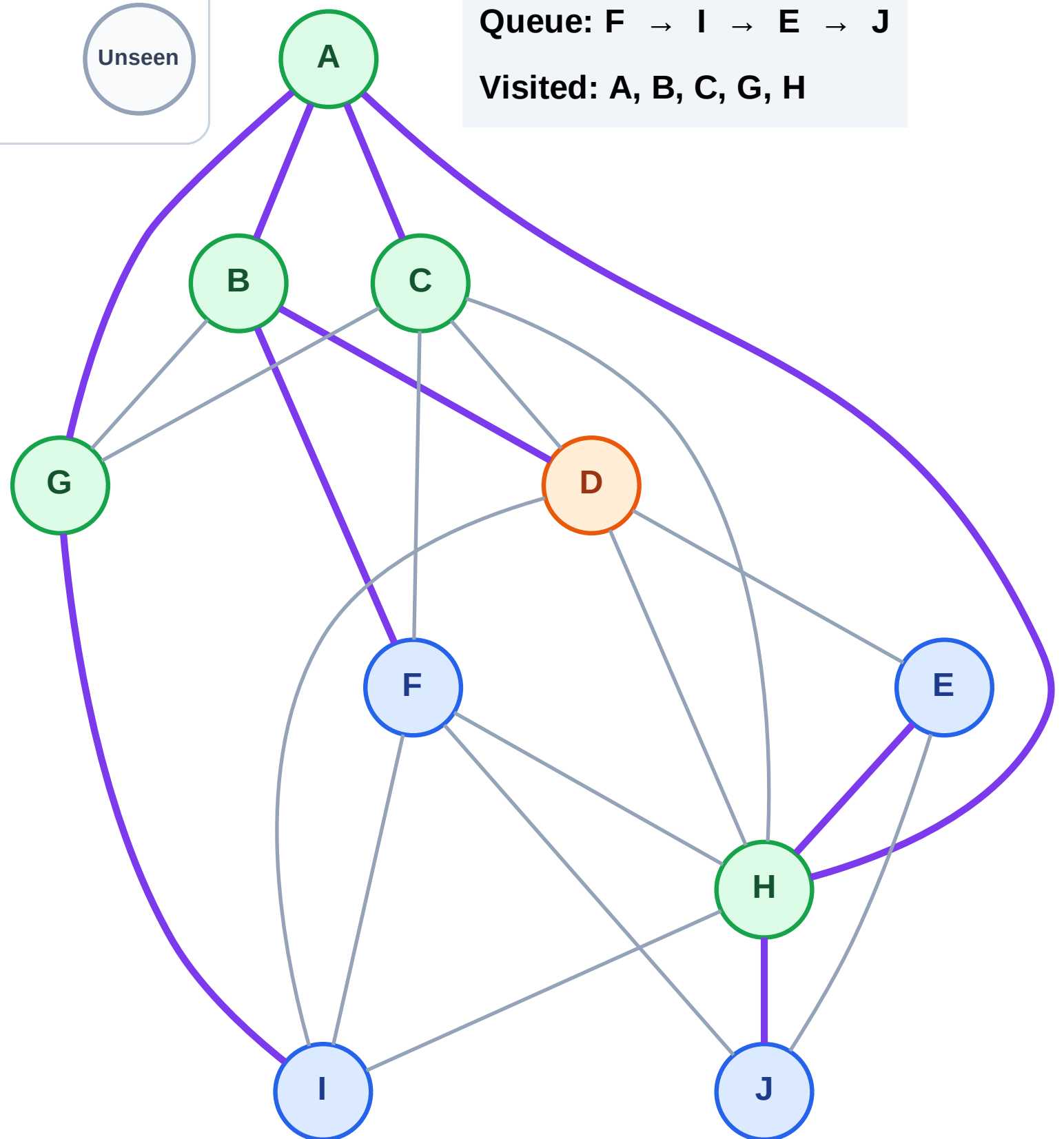
Step 12: Process node D

Legend



Queue: F → I → E → J

Visited: A, B, C, G, H



BFS Traversal

Step 13: No new neighbors from D

Legend

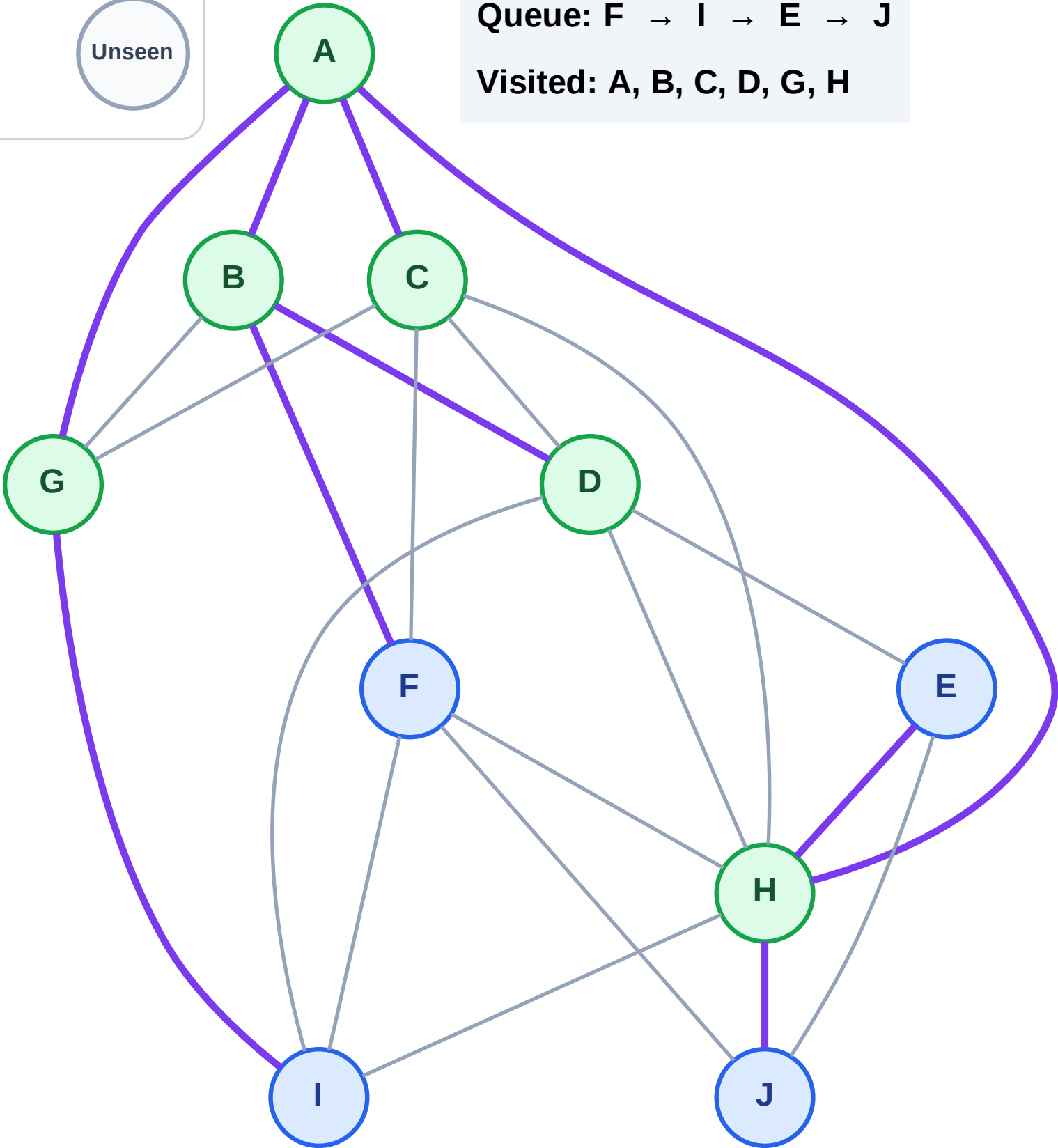
Complete

Processing

Discovered

Unseen

Queue: F → I → E → J
Visited: A, B, C, D, G, H



BFS Traversal

Step 14: Process node F

Legend

Complete

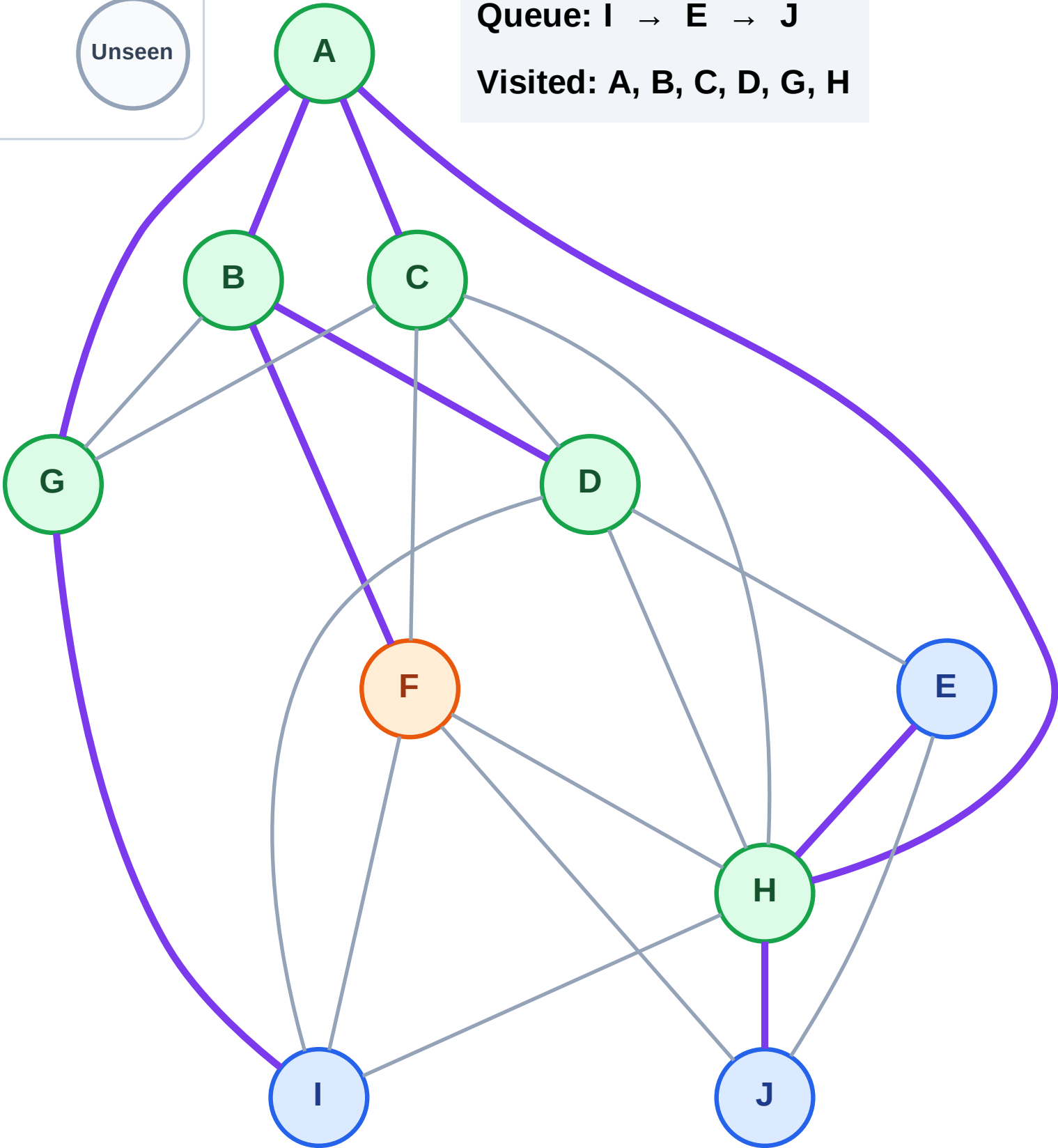
Processing

Discovered

Unseen

Queue: I → E → J

Visited: A, B, C, D, G, H



BFS Traversal

Step 15: No new neighbors from F

Legend

Complete

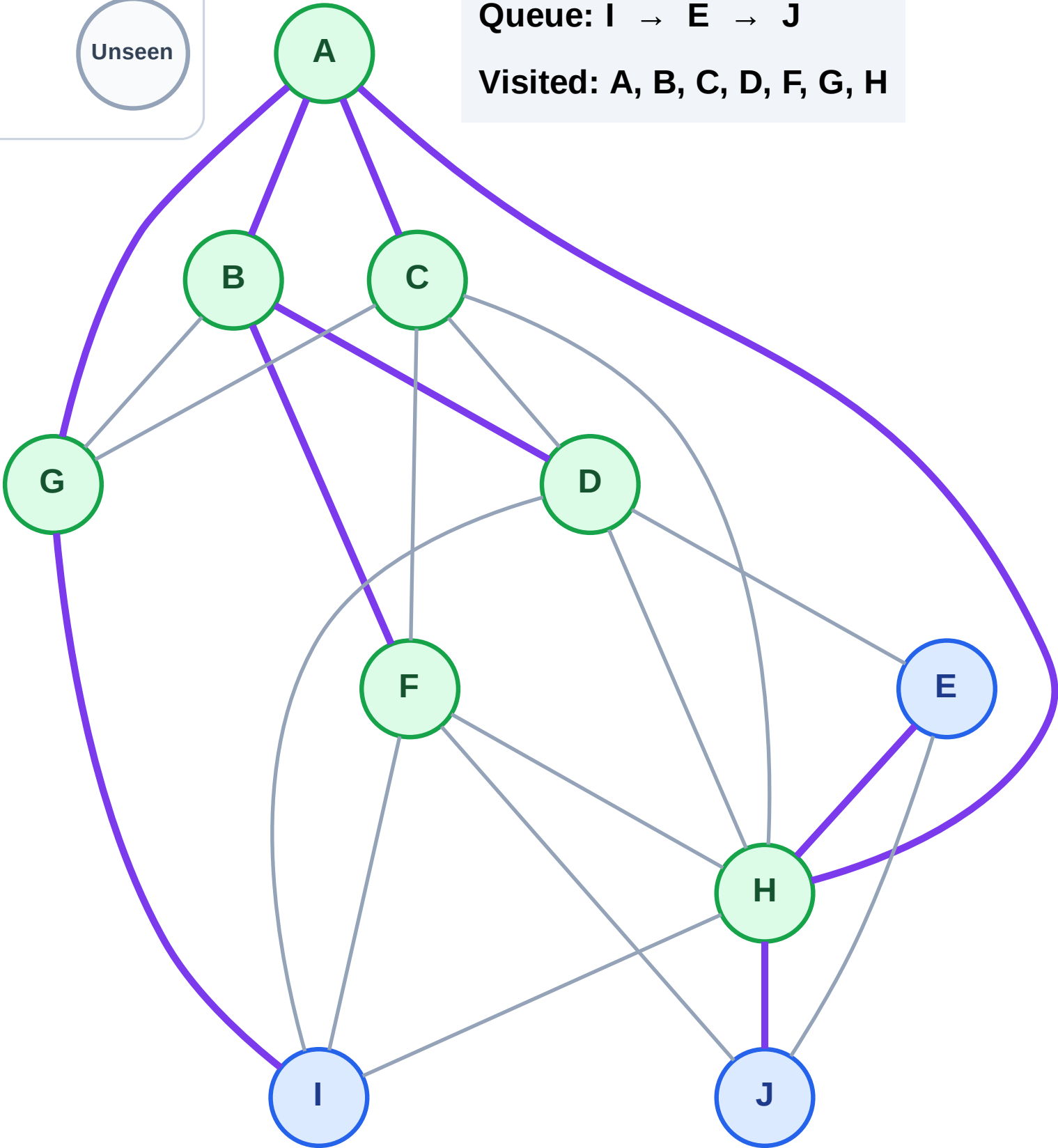
Processing

Discovered

Unseen

Queue: I → E → J

Visited: A, B, C, D, F, G, H



BFS Traversal

Step 16: Process node I

Legend

Complete

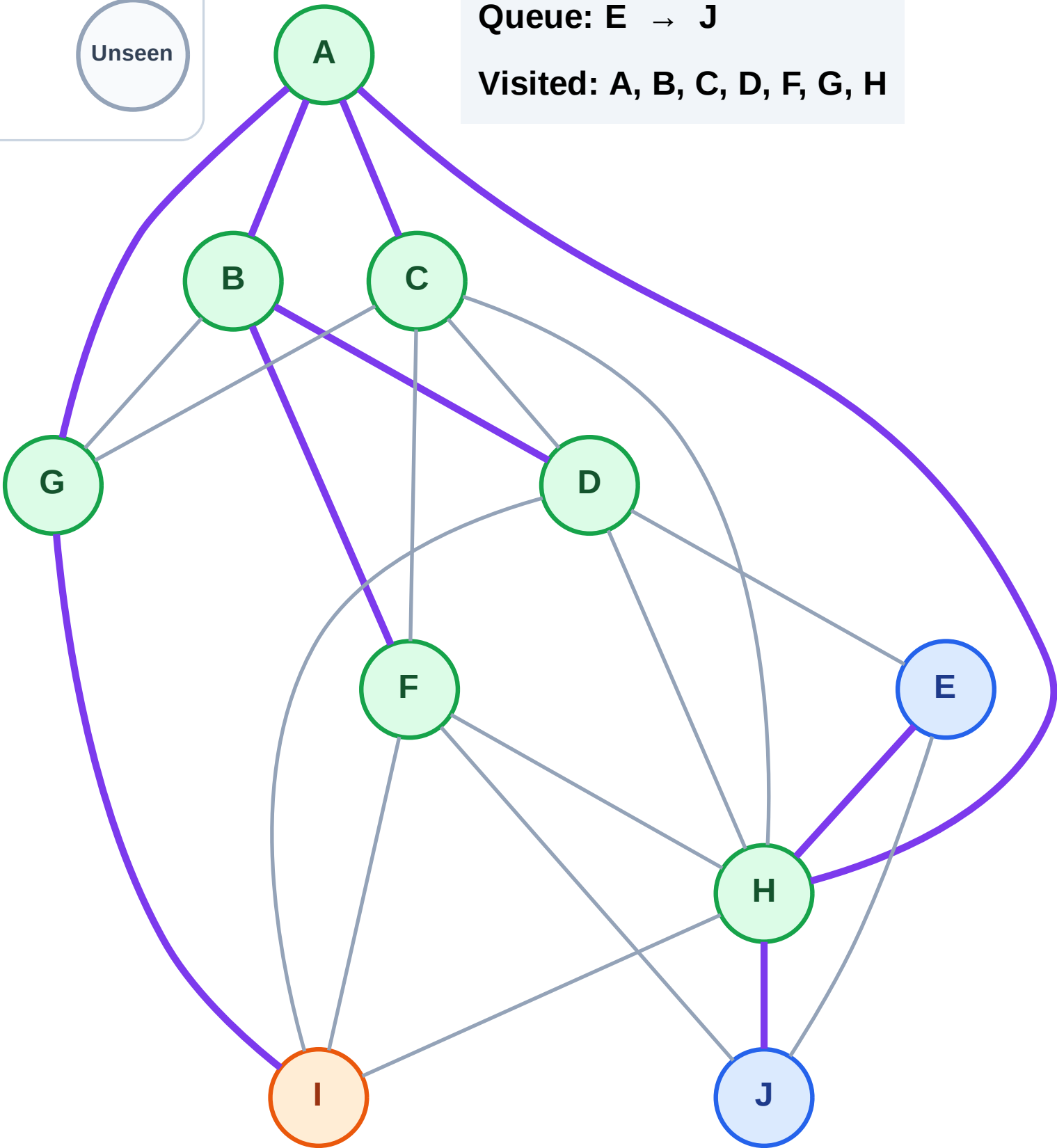
Processing

Discovered

Unseen

Queue: E → J

Visited: A, B, C, D, F, G, H



BFS Traversal

Step 17: No new neighbors from I

Legend

Complete

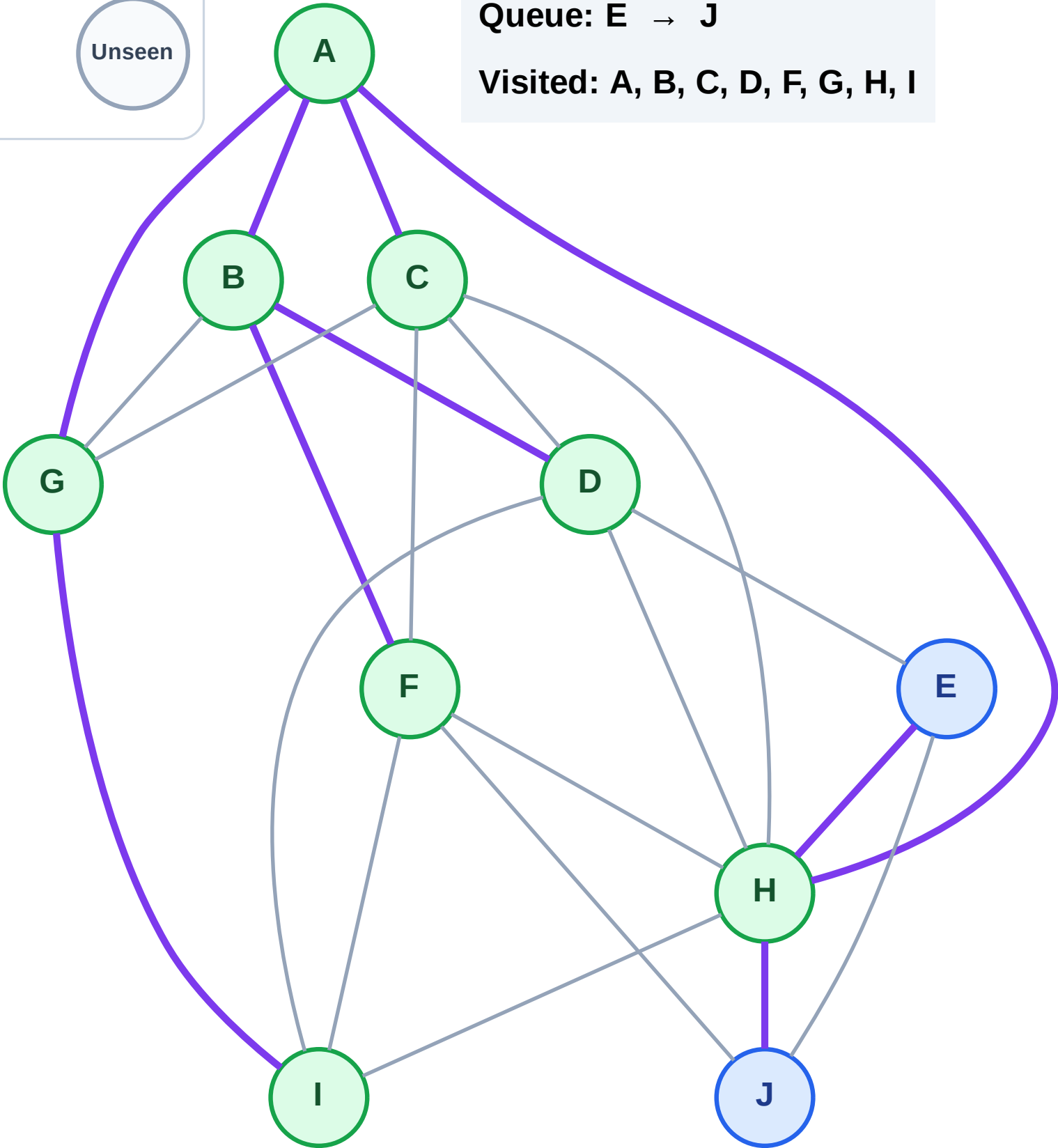
Processing

Discovered

Unseen

Queue: E → J

Visited: A, B, C, D, F, G, H, I



BFS Traversal

Step 18: Process node E

Legend

Complete

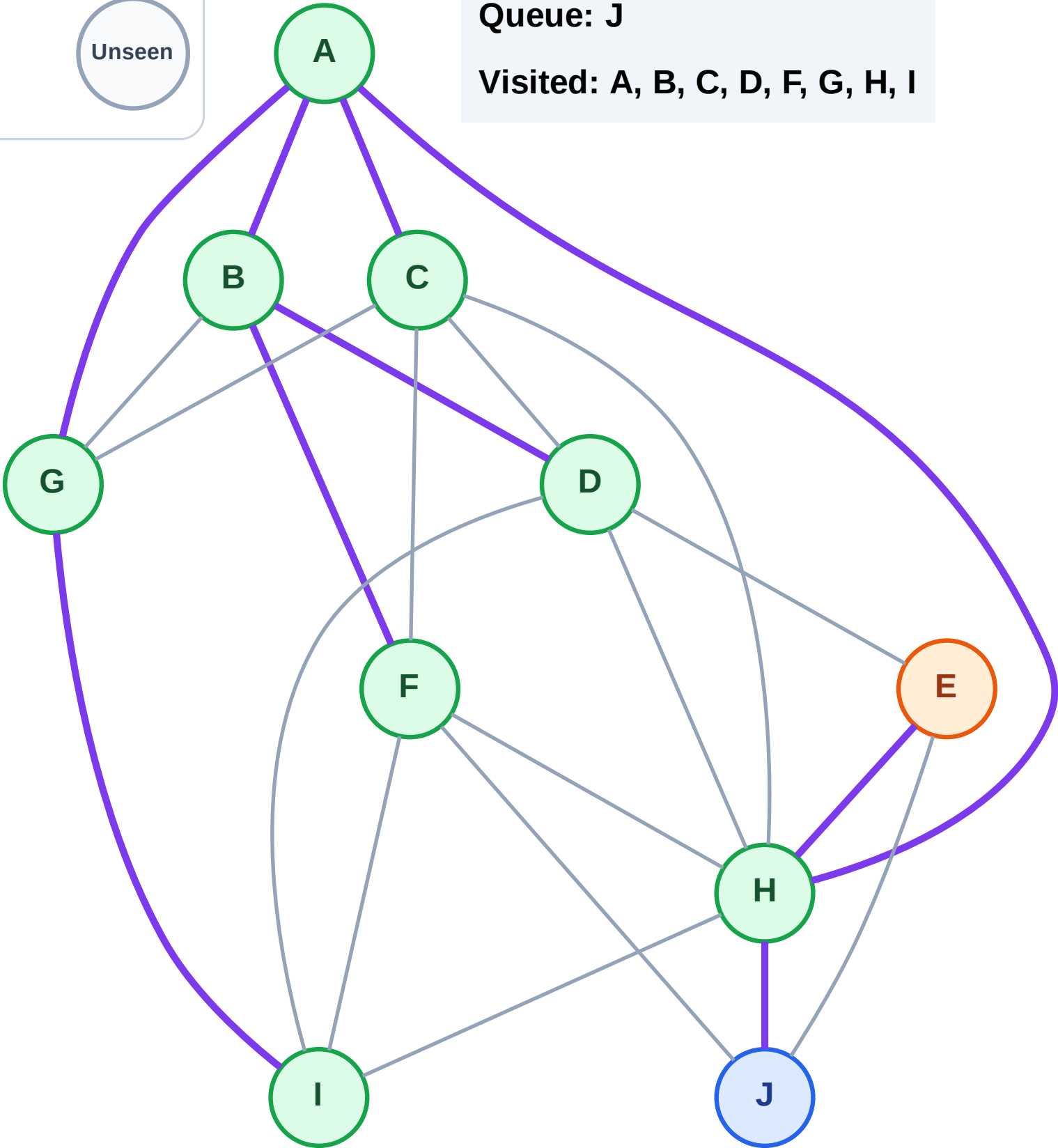
Processing

Discovered

Unseen

Queue: J

Visited: A, B, C, D, F, G, H, I



BFS Traversal

Step 19: No new neighbors from E

Legend

Complete

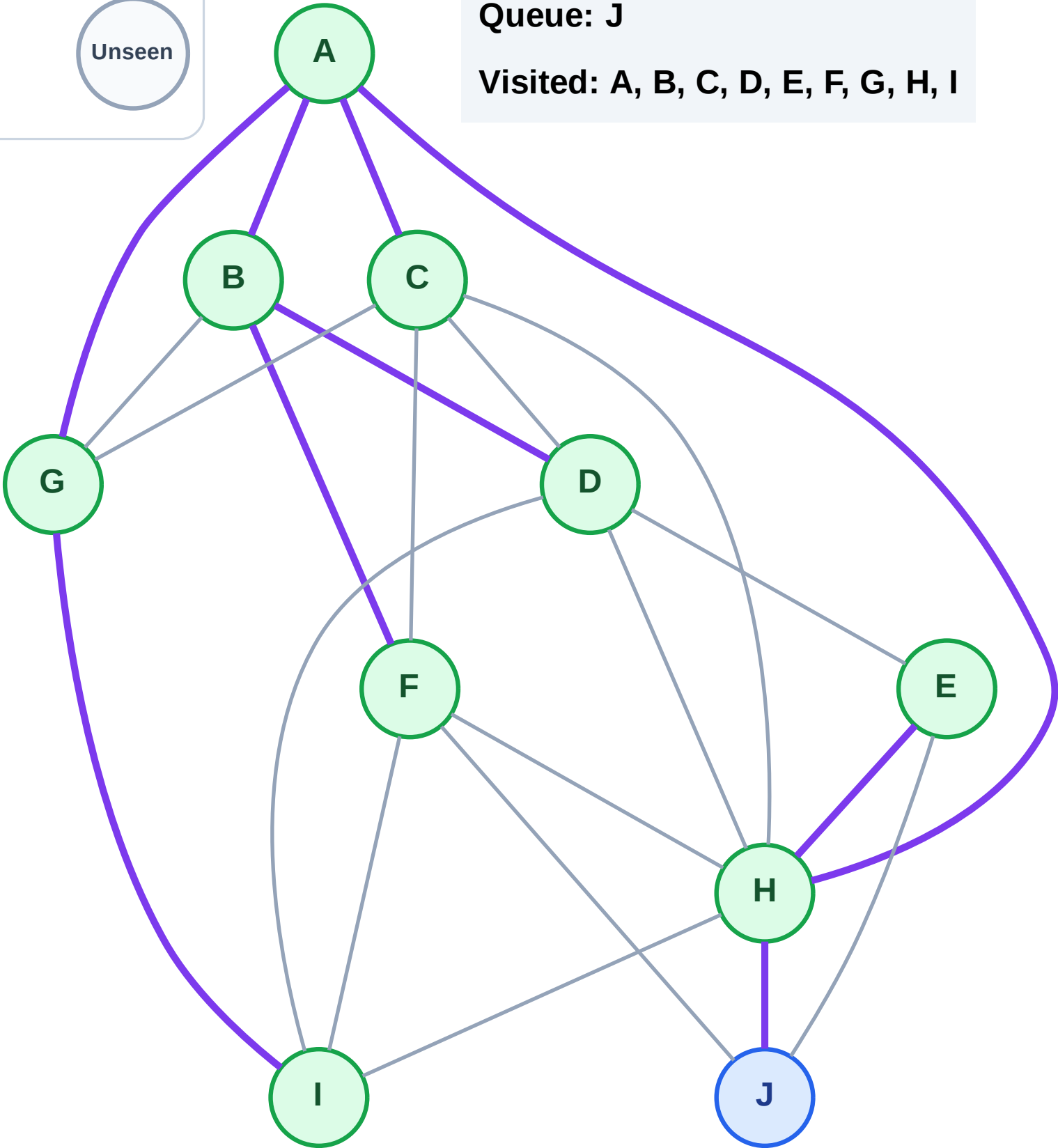
Processing

Discovered

Unseen

Queue: J

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

Step 20: Process node J

Legend

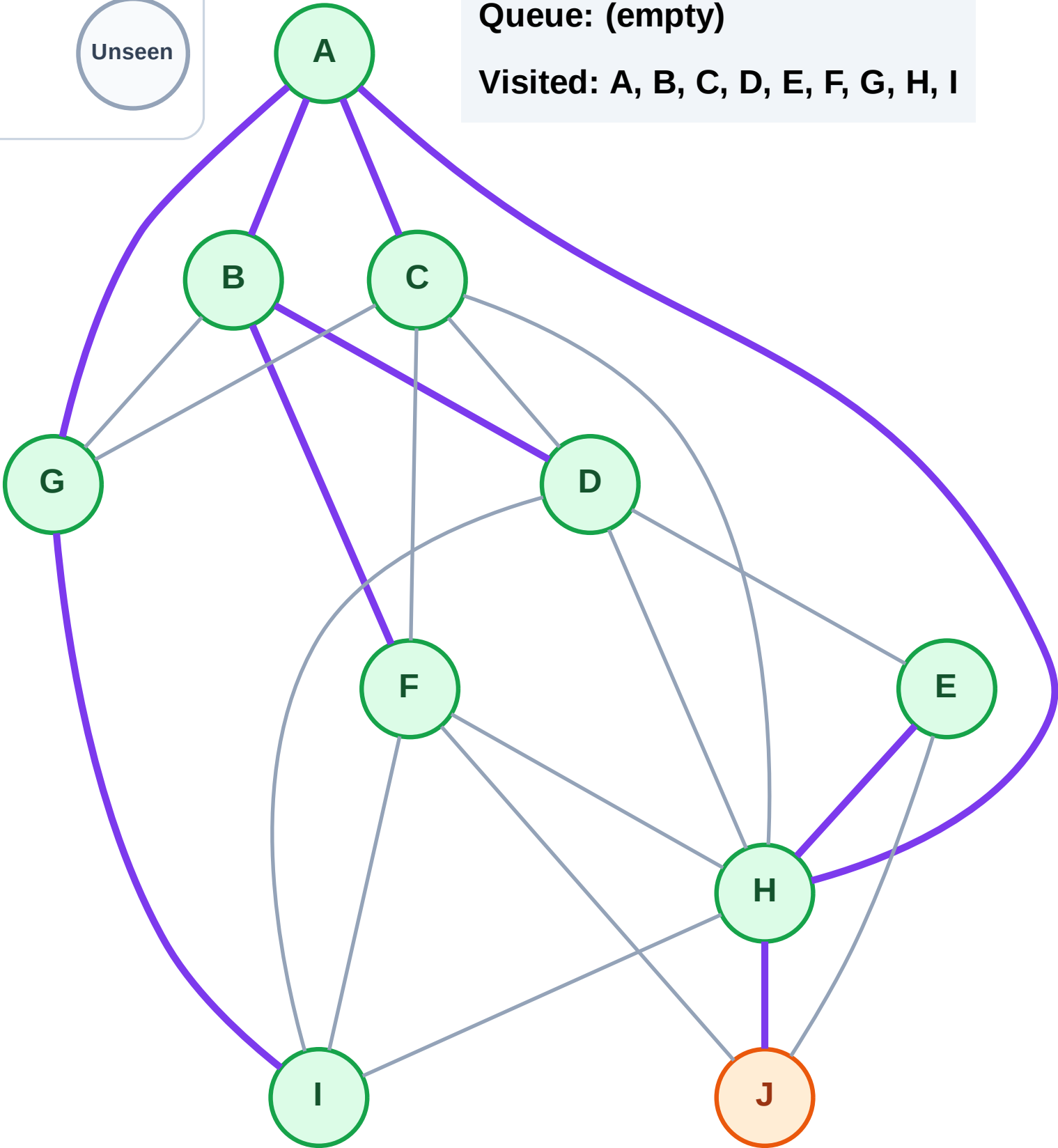
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: A, B, C, D, E, F, G, H, I

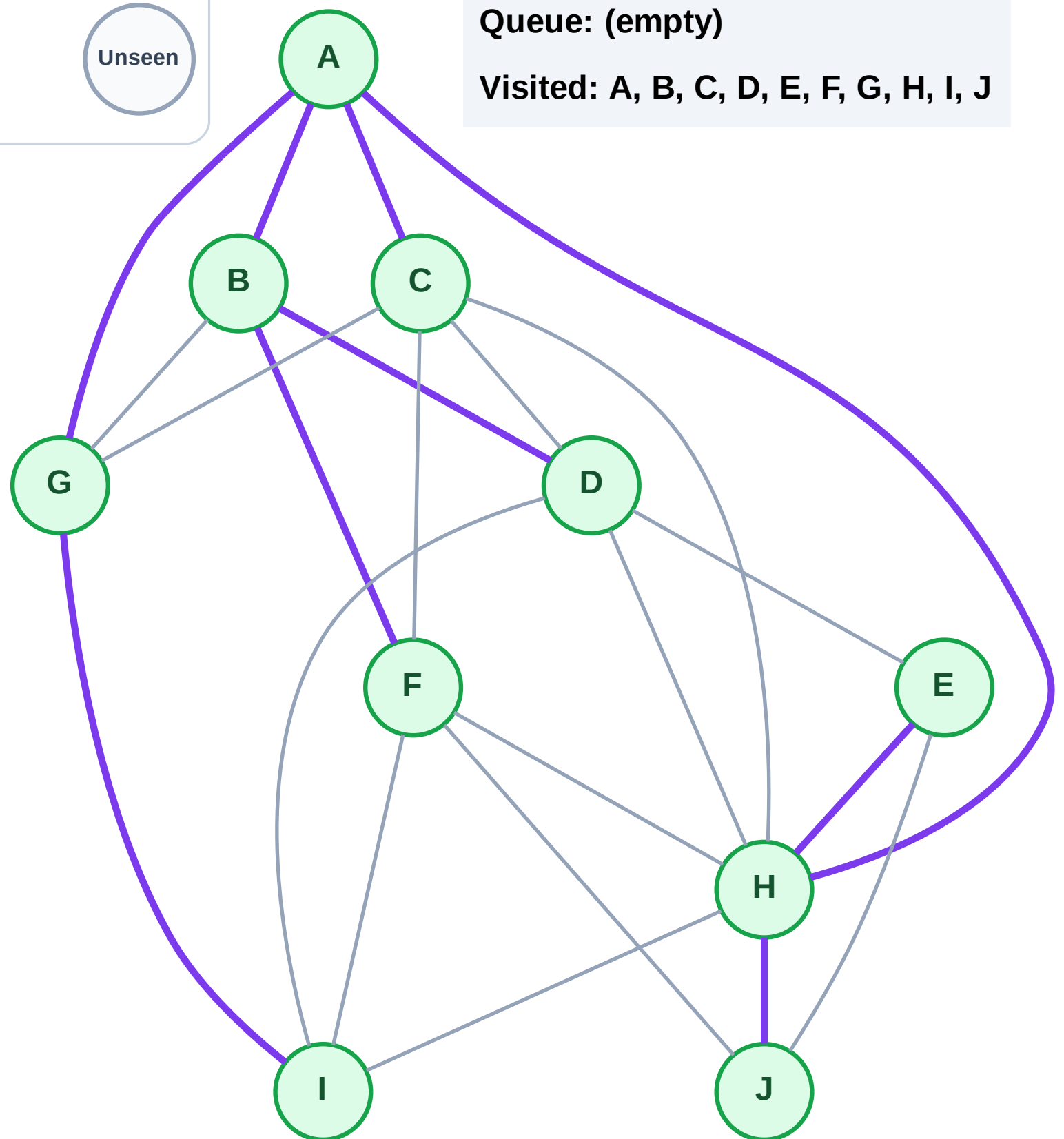


Step 21: No new neighbors from J

Legend

Unseen

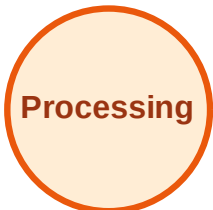
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

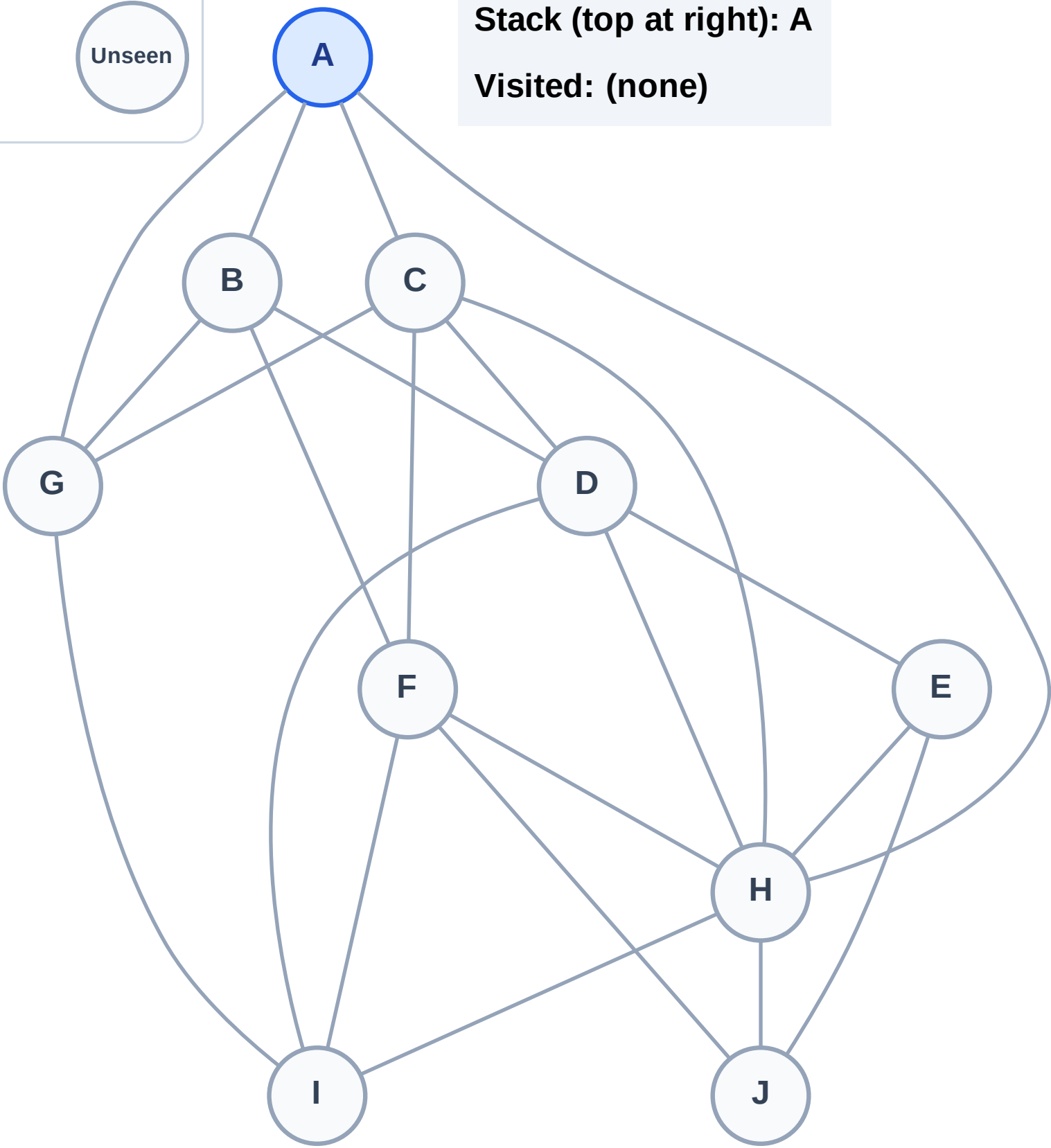
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

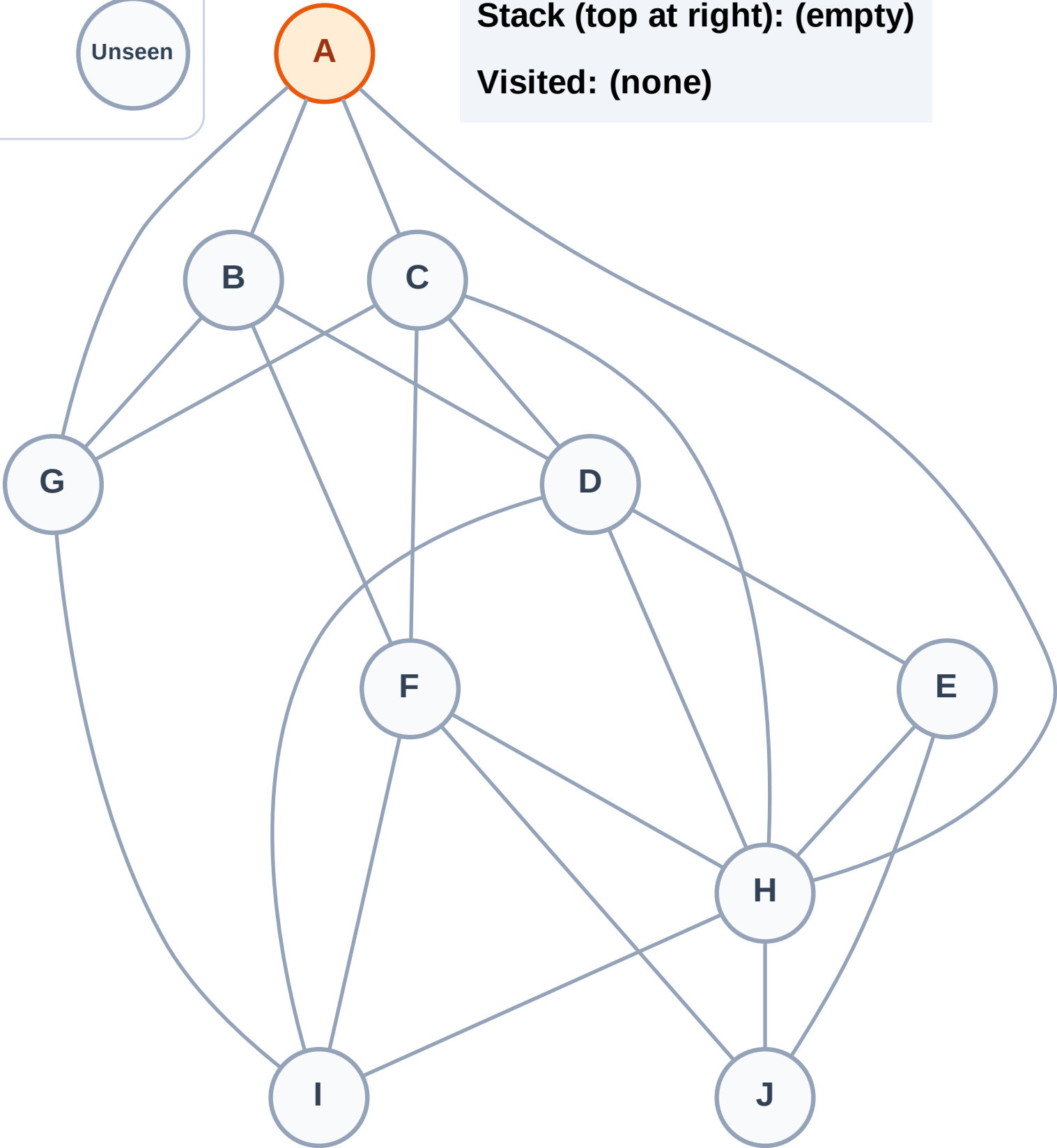
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover H, G, C, B from A

Legend

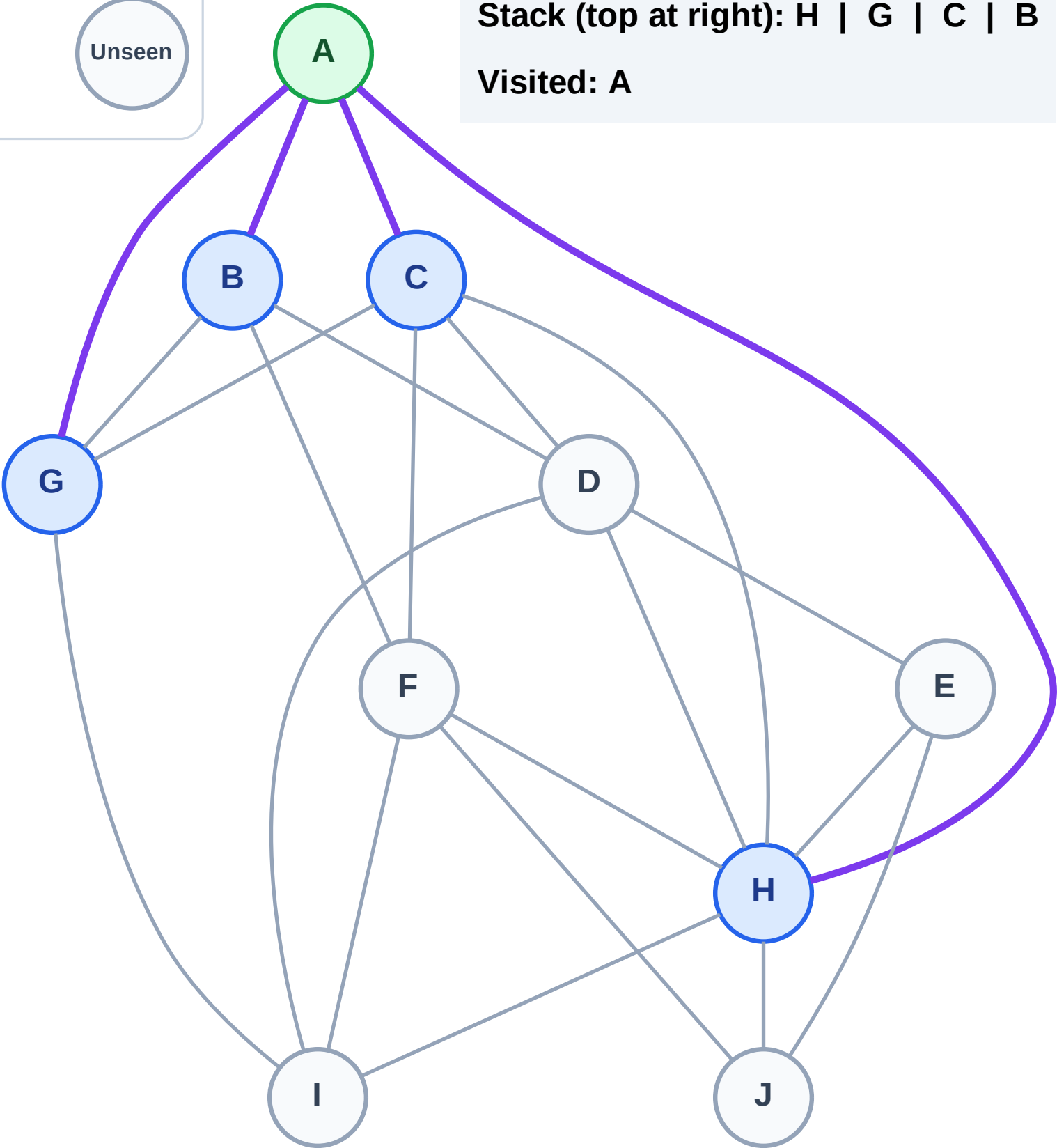
Complete

Processing

Discovered

Unseen

Stack (top at right): H | G | C | B
Visited: A



DFS Traversal

Step 4: Process node B

Legend

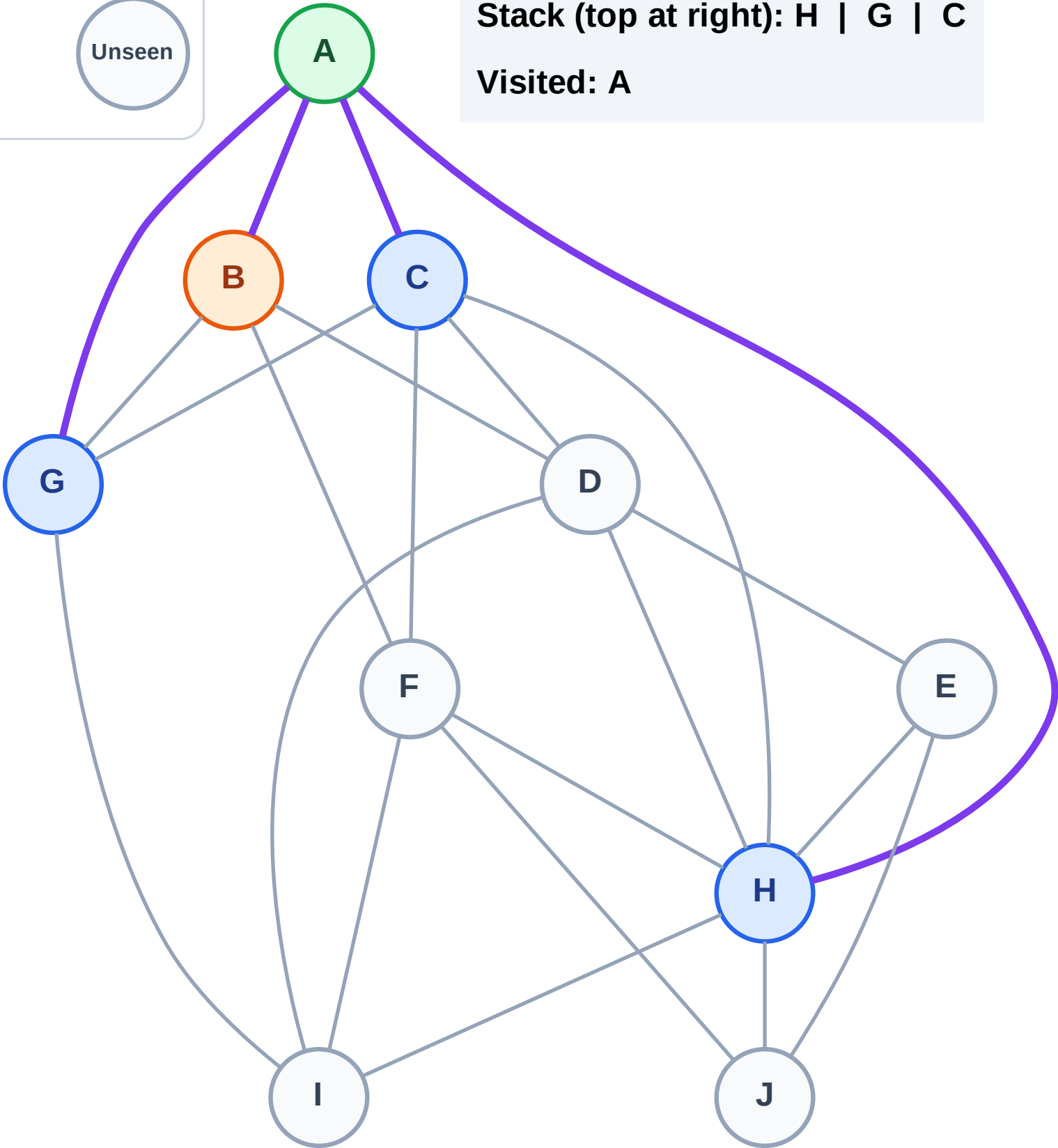
Complete

Processing

Discovered

Unseen

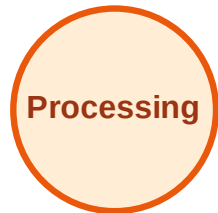
Stack (top at right): H | G | C
Visited: A



DFS Traversal

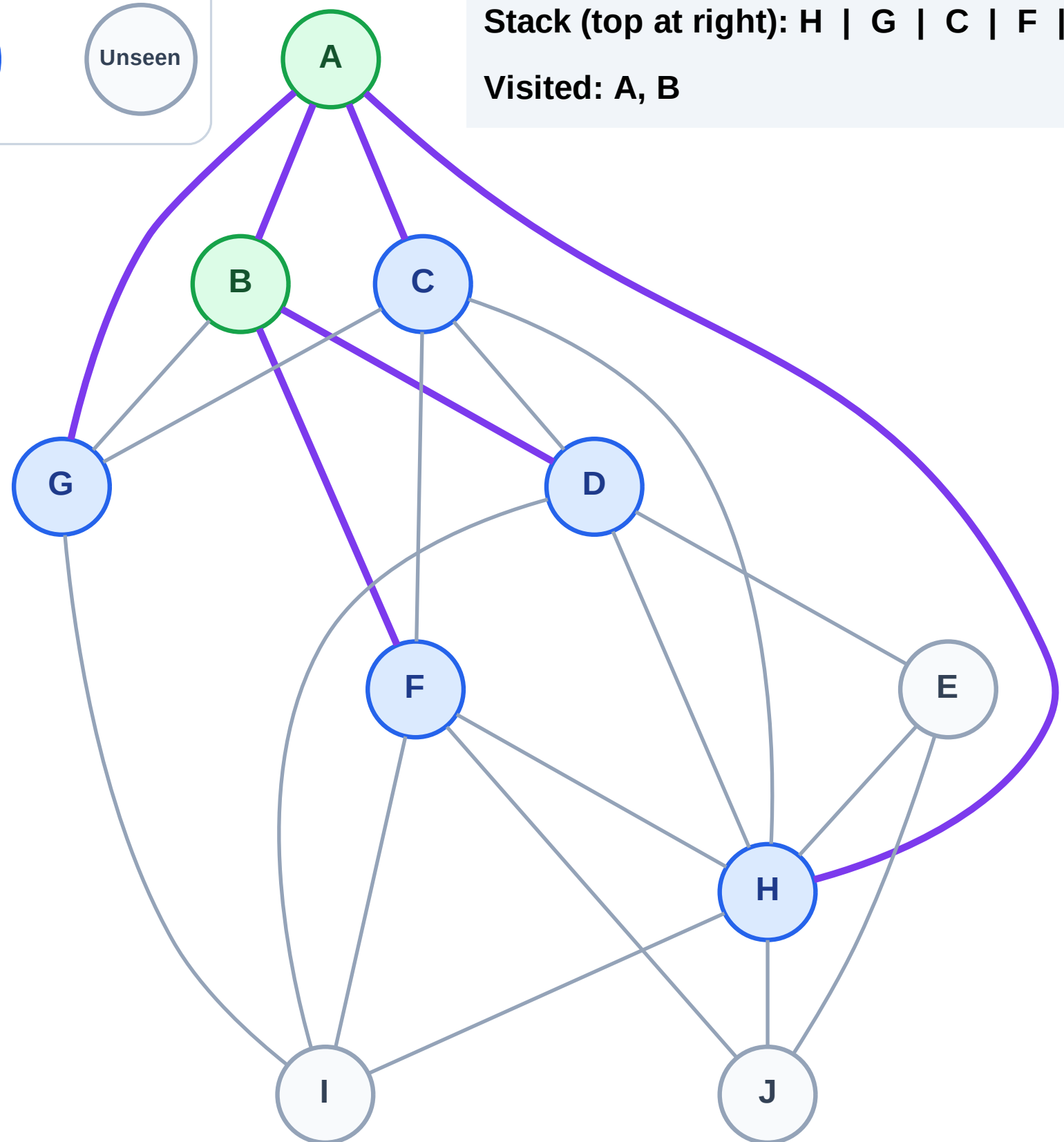
Step 5: Discover F, D from B

Legend



Stack (top at right): H | G | C | F | D

Visited: A, B



DFS Traversal

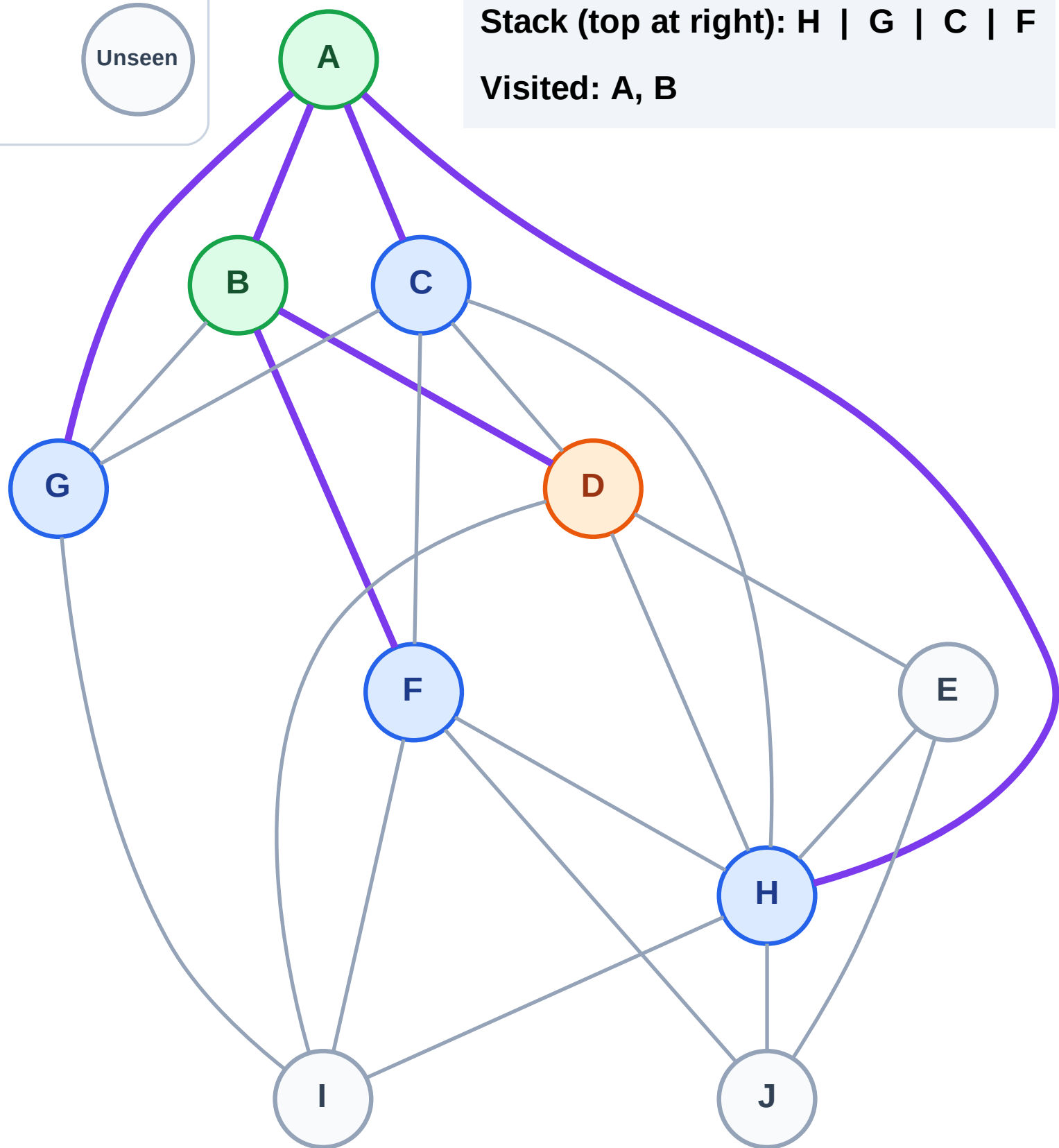
Step 6: Process node D

Legend



Stack (top at right): H | G | C | F

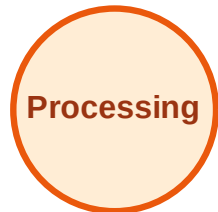
Visited: A, B



DFS Traversal

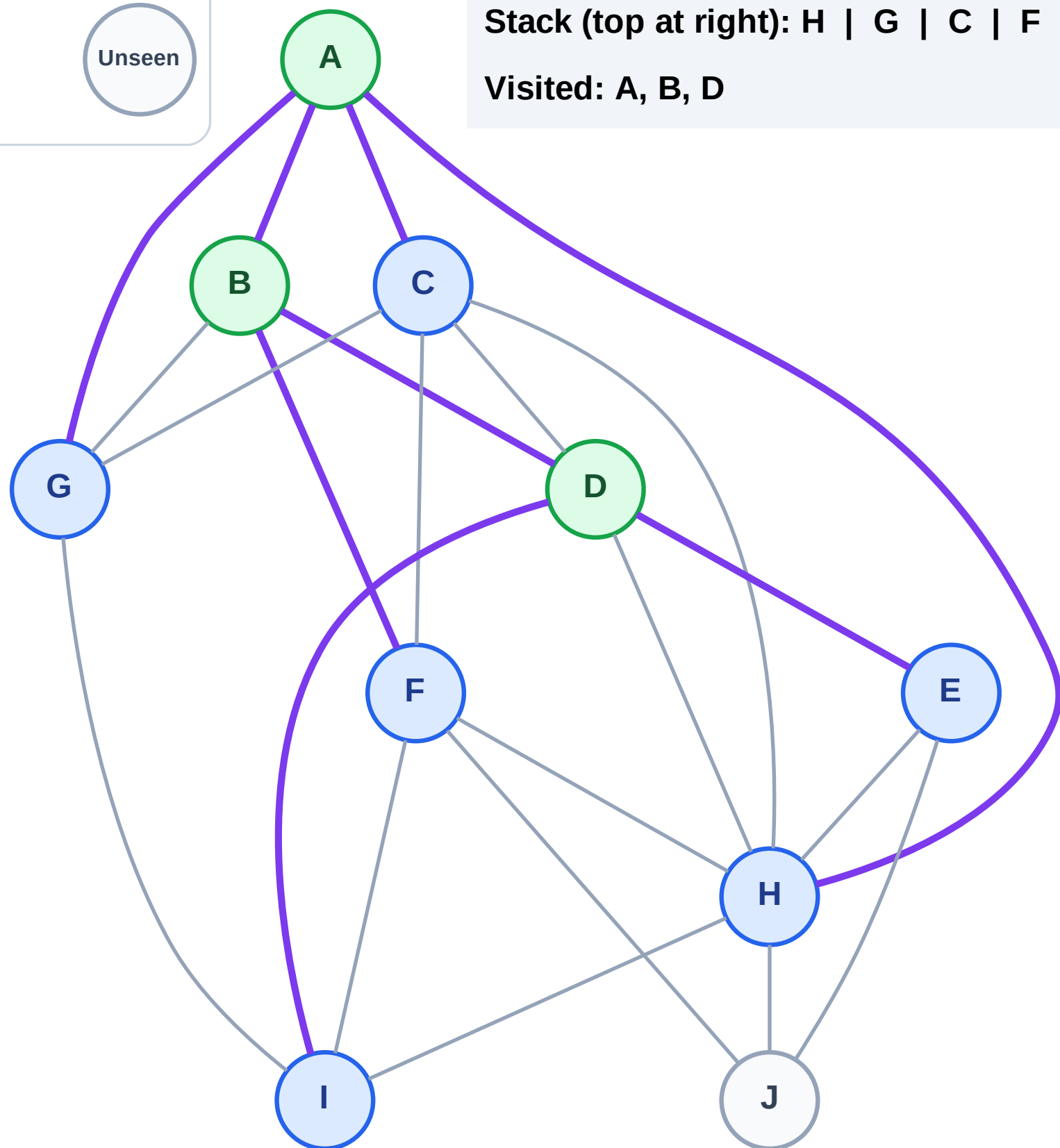
Step 7: Discover I, E from D

Legend



Stack (top at right): H | G | C | F | I | E

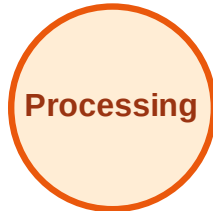
Visited: A, B, D



DFS Traversal

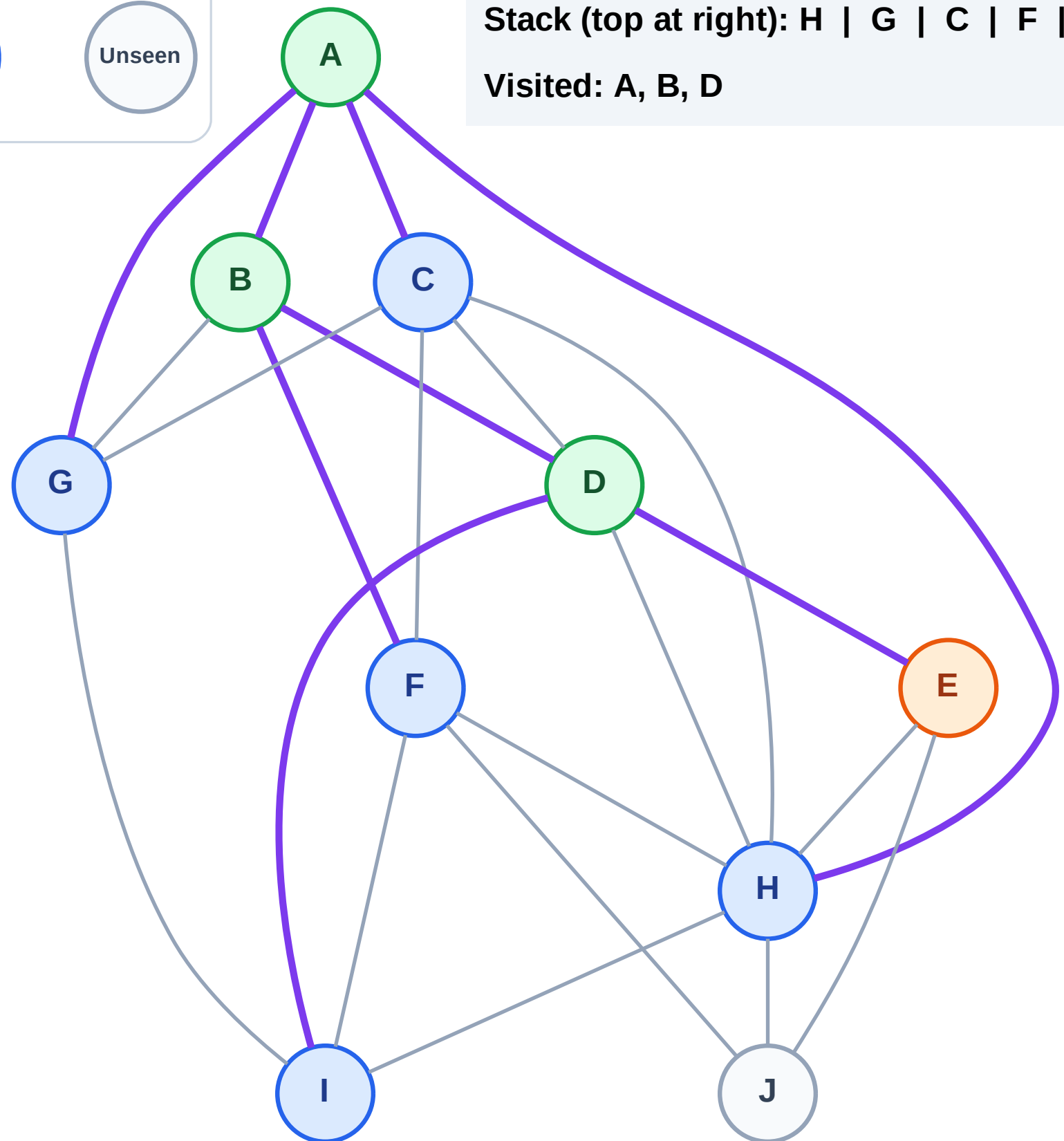
Step 8: Process node E

Legend



Stack (top at right): H | G | C | F | I

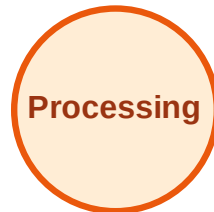
Visited: A, B, D



DFS Traversal

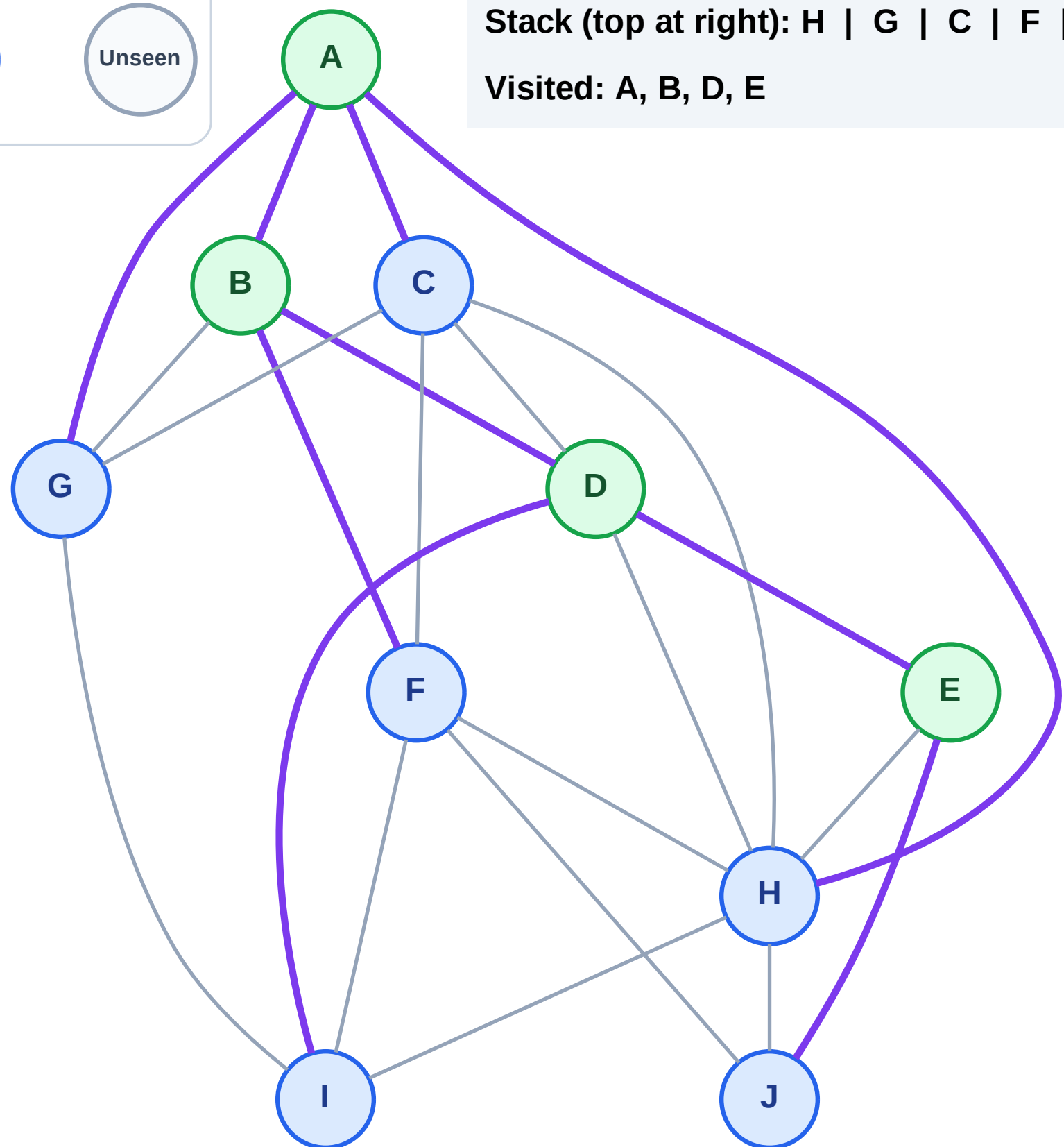
Step 9: Discover J from E

Legend



Stack (top at right): H | G | C | F | I | J

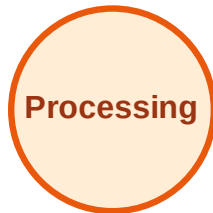
Visited: A, B, D, E



DFS Traversal

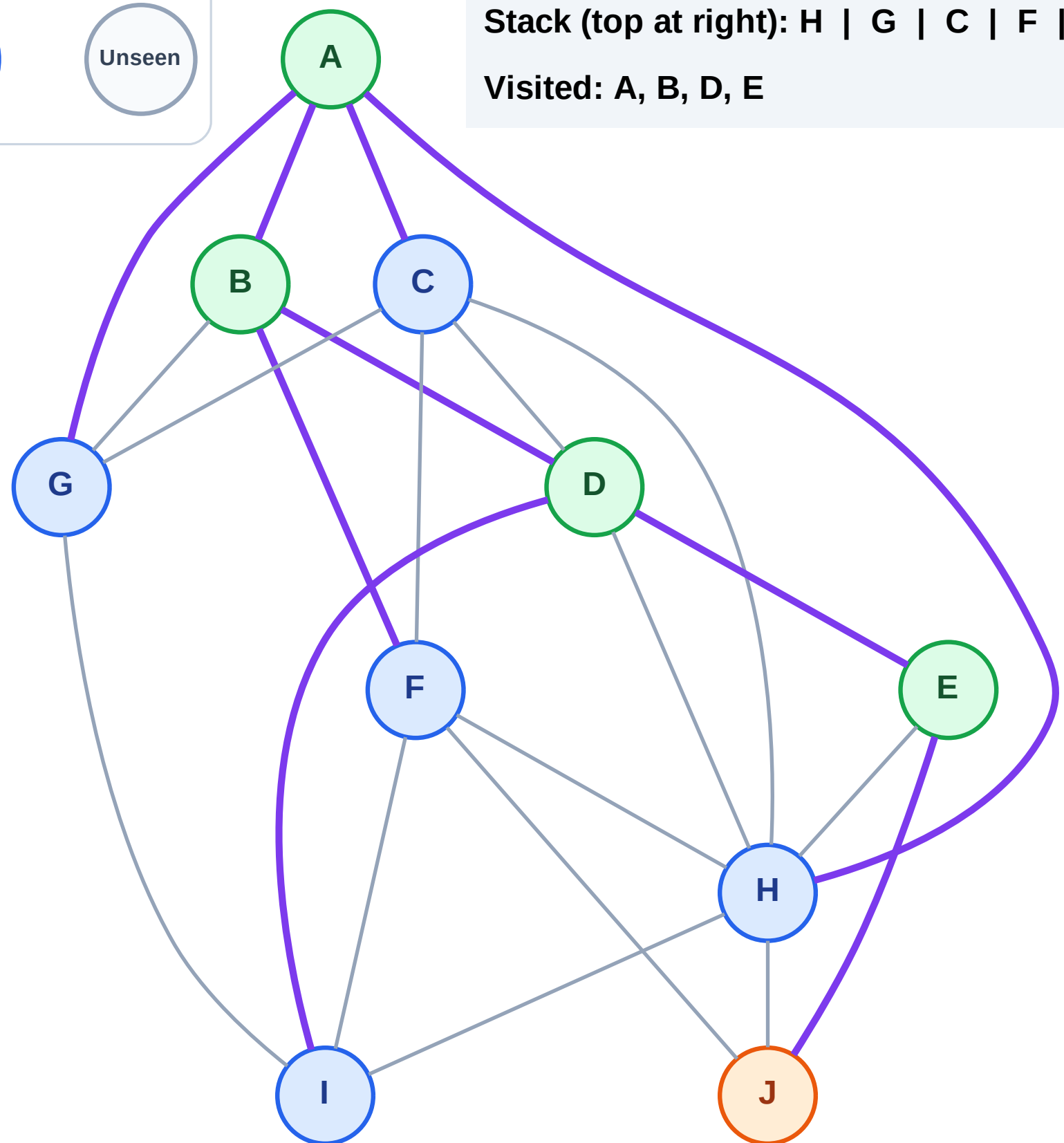
Step 10: Process node J

Legend



Stack (top at right): H | G | C | F | I

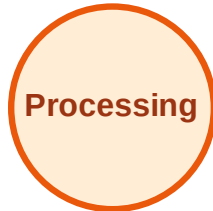
Visited: A, B, D, E



DFS Traversal

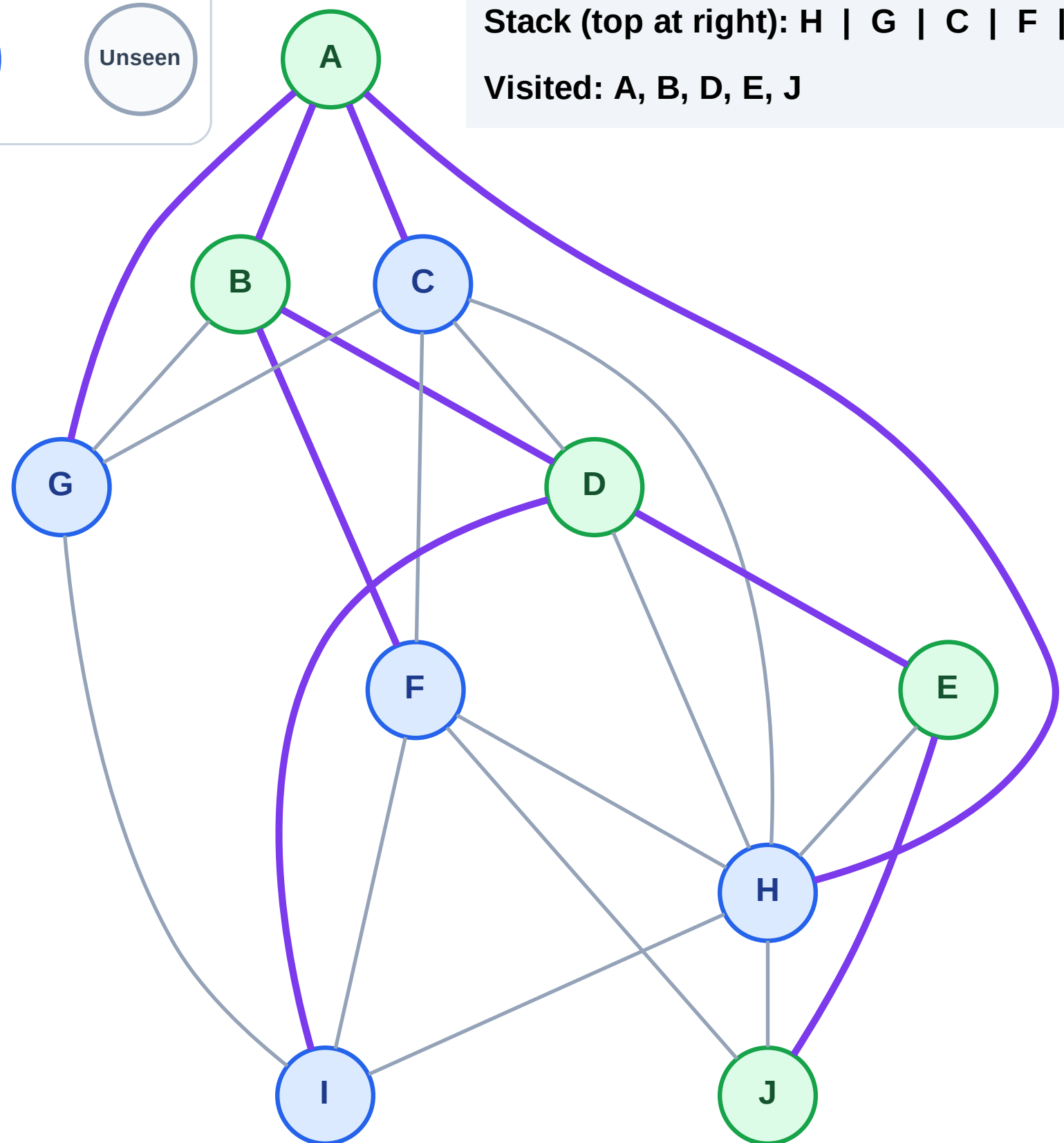
Step 11: No new neighbors from J

Legend



Stack (top at right): H | G | C | F | I

Visited: A, B, D, E, J



DFS Traversal

Step 12: Process node I

Legend

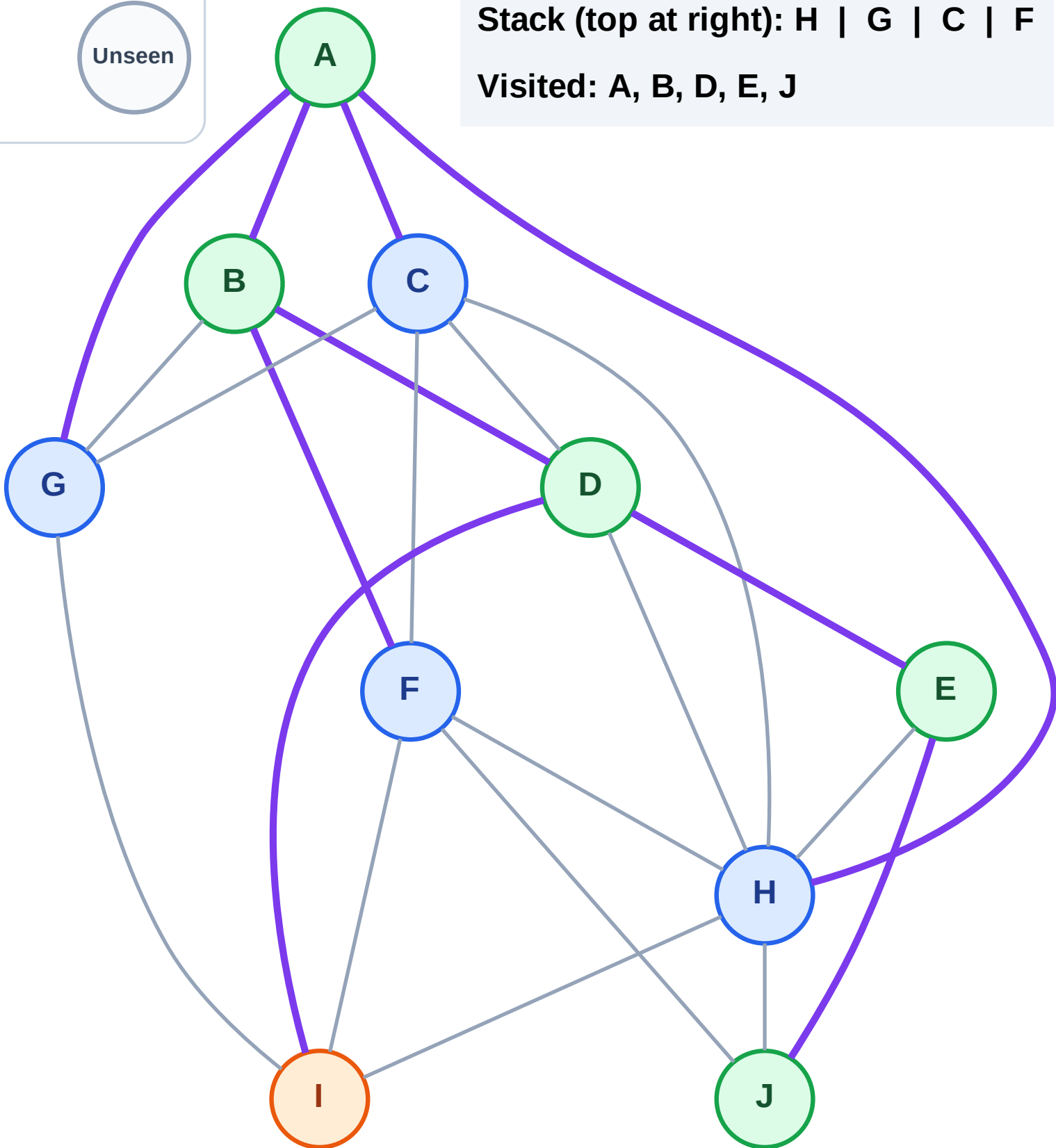
Complete

Processing

Discovered

Unseen

Stack (top at right): H | G | C | F
Visited: A, B, D, E, J



DFS Traversal

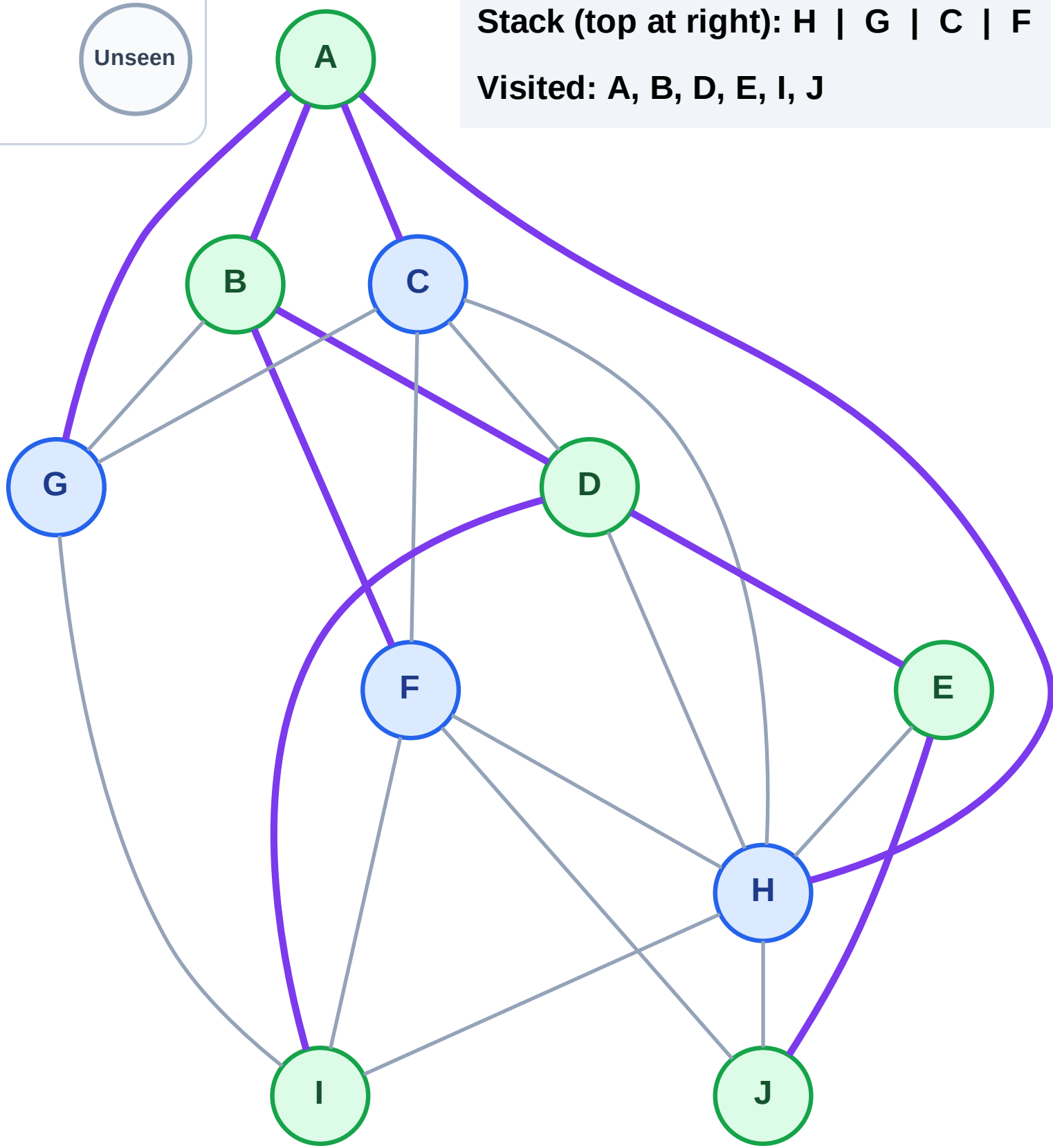
Step 13: No new neighbors from I

Legend



Stack (top at right): H | G | C | F

Visited: A, B, D, E, I, J



DFS Traversal

Step 14: Process node F

Legend

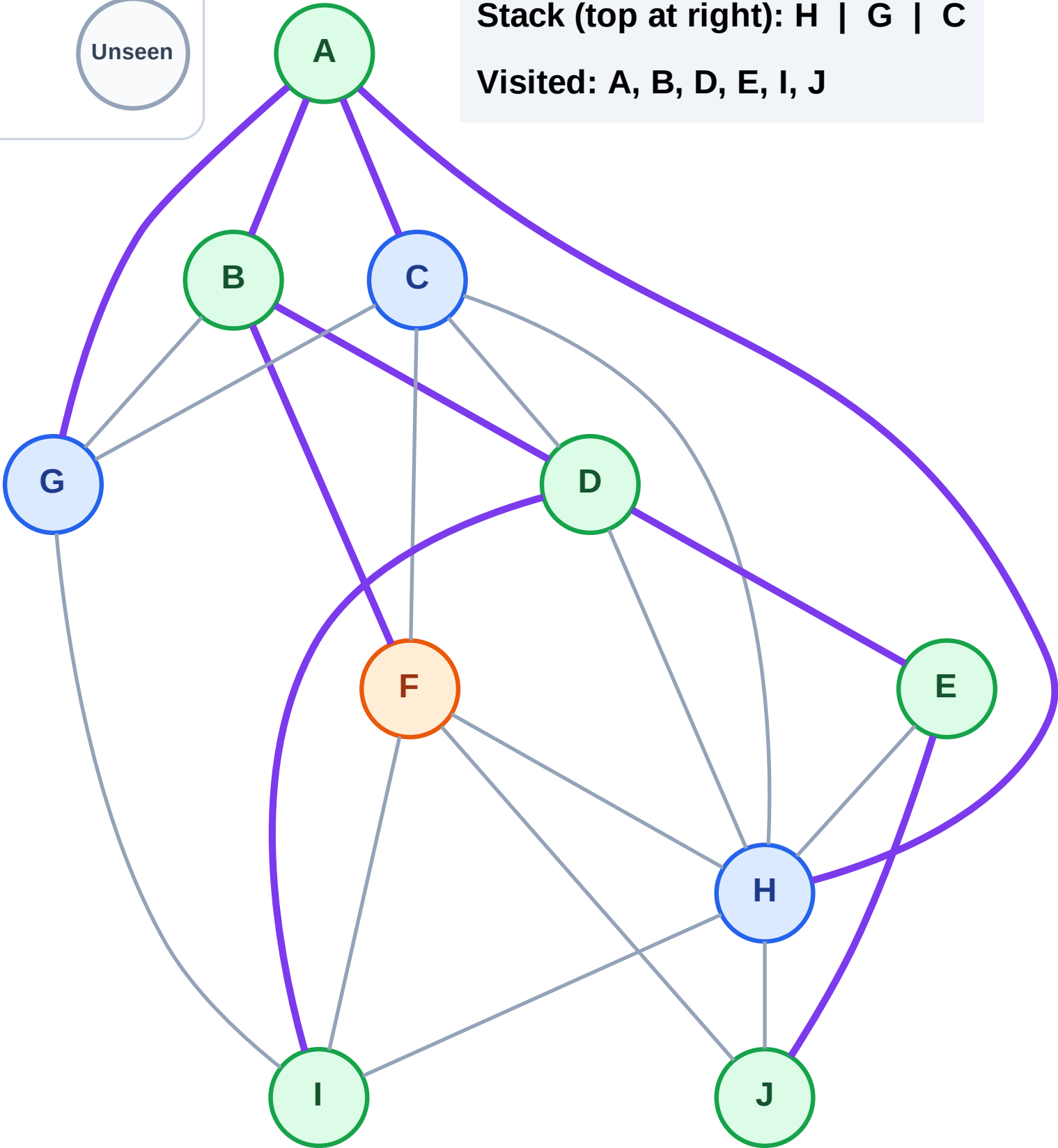
Complete

Processing

Discovered

Unseen

Stack (top at right): H | G | C
Visited: A, B, D, E, I, J



DFS Traversal

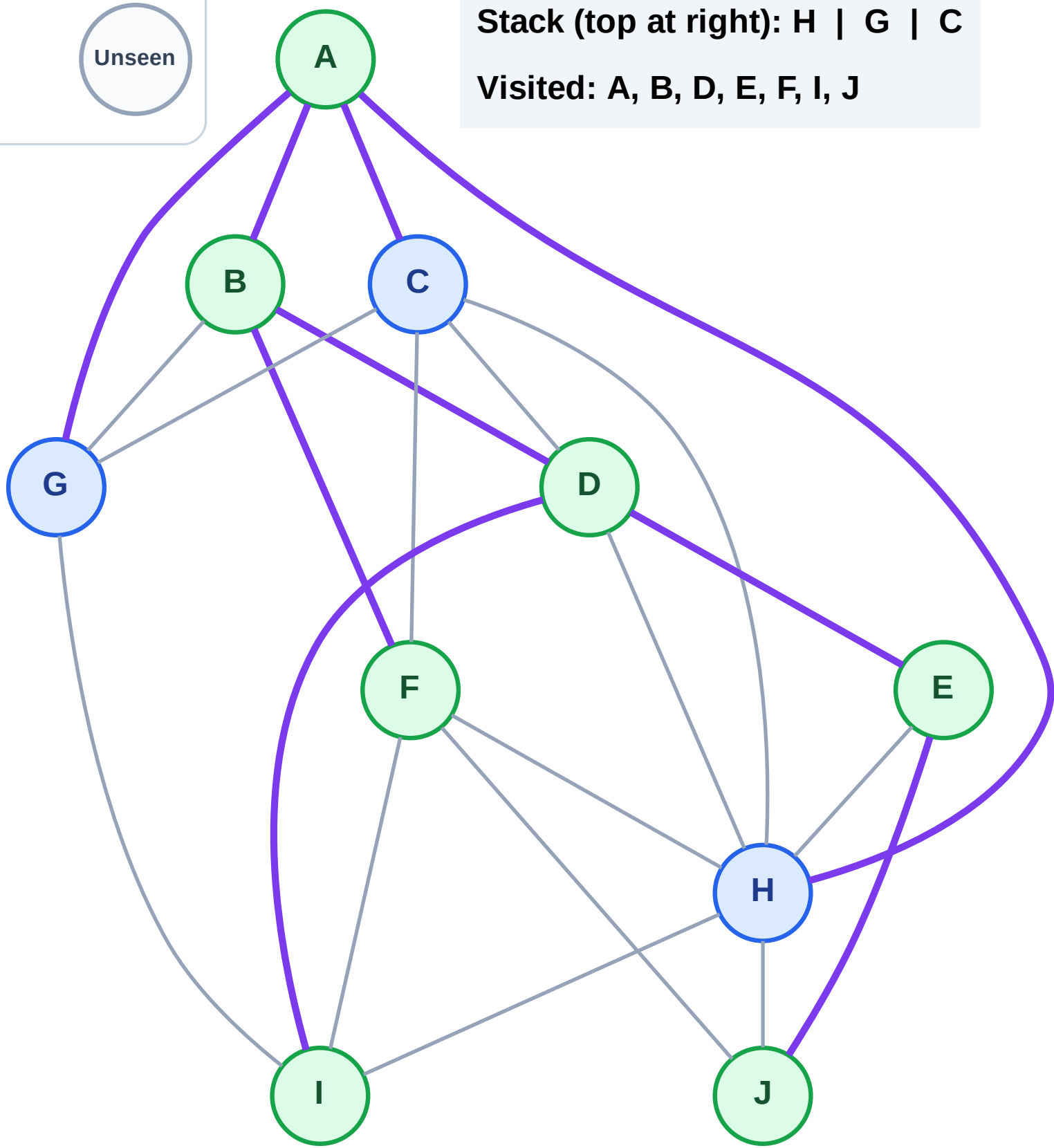
Step 15: No new neighbors from F

Legend



Stack (top at right): H | G | C

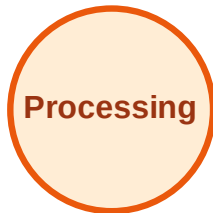
Visited: A, B, D, E, F, I, J



DFS Traversal

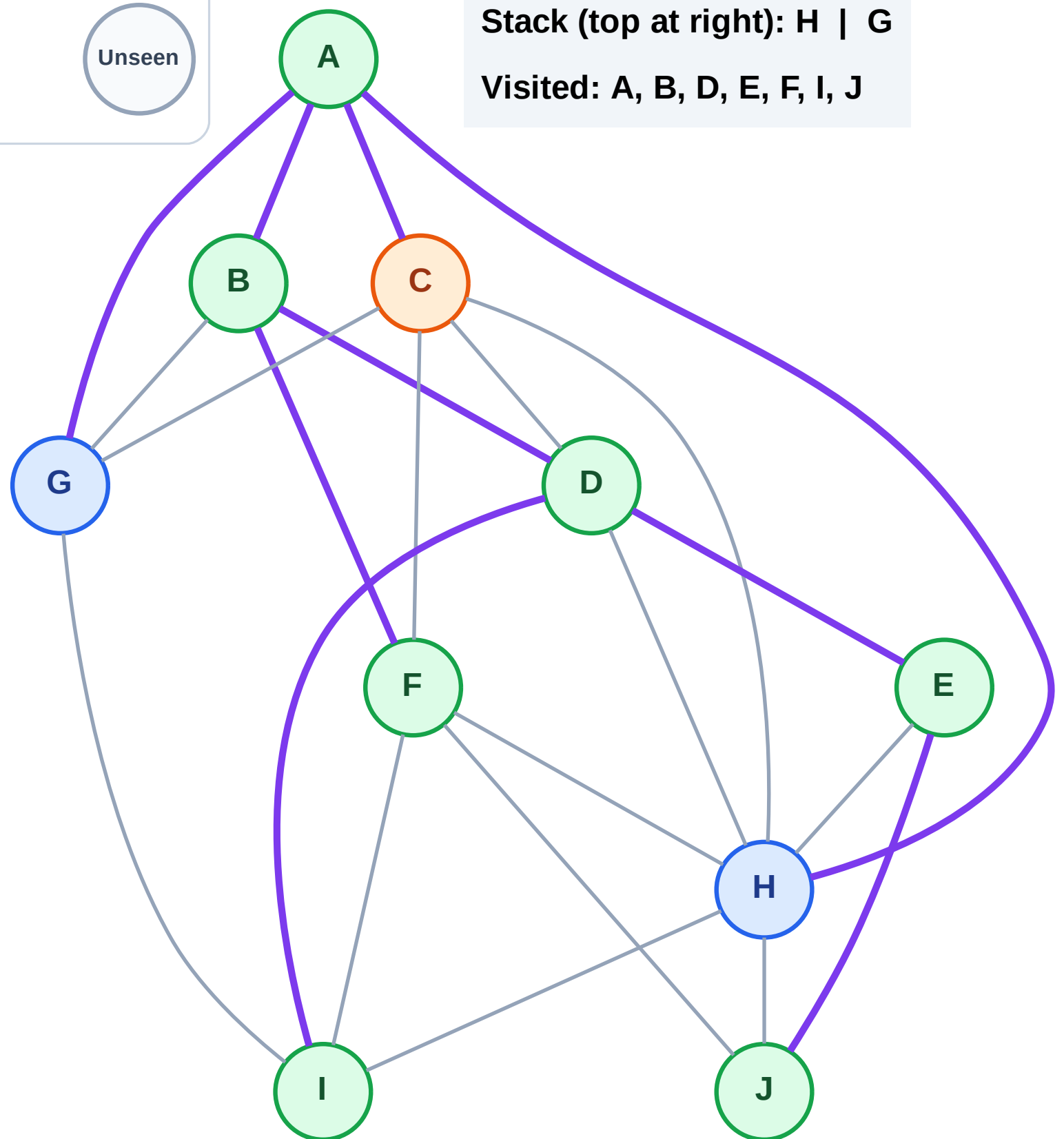
Step 16: Process node C

Legend



Stack (top at right): H | G

Visited: A, B, D, E, F, I, J



DFS Traversal

Step 17: No new neighbors from C

Legend

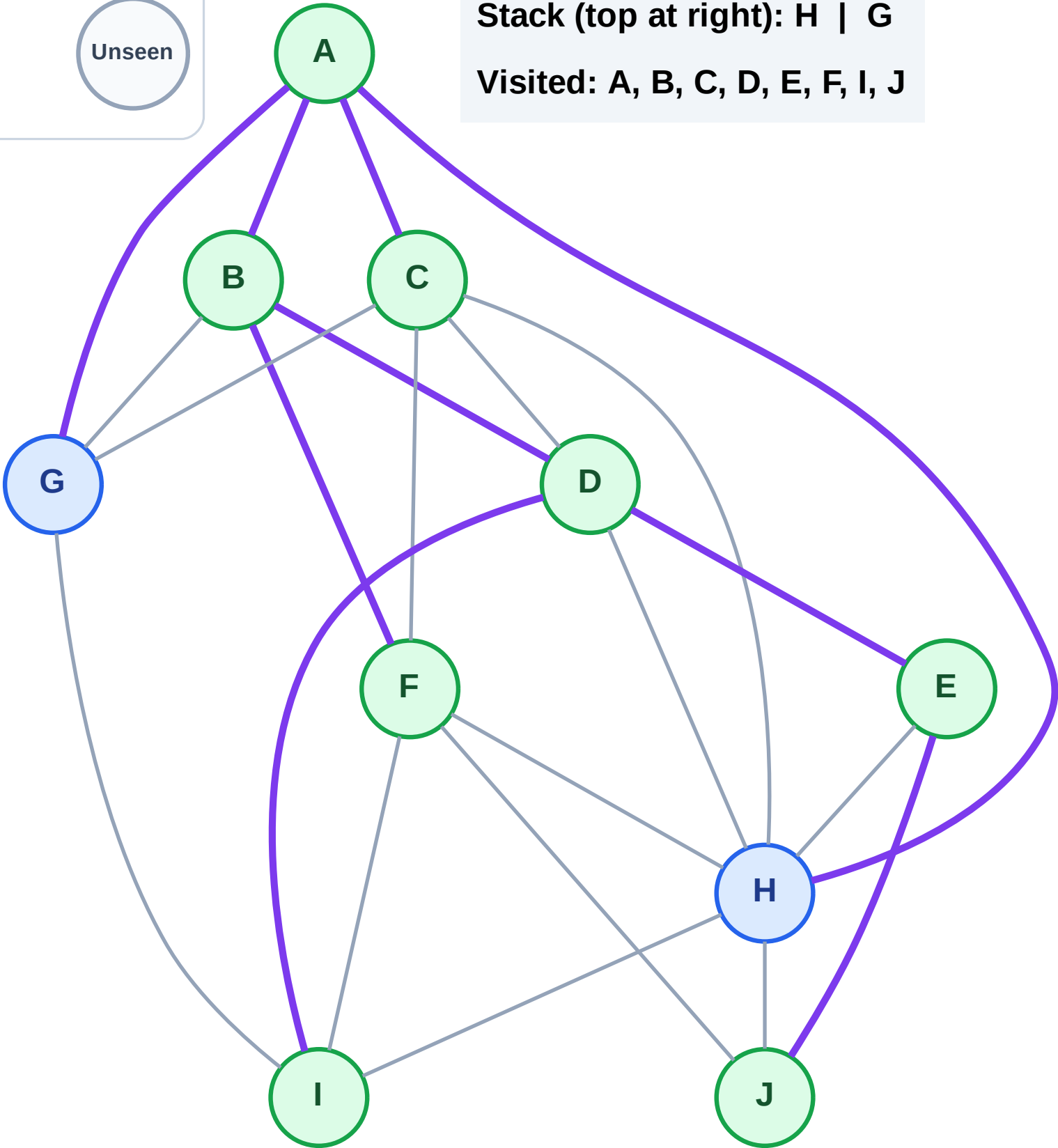
Complete

Processing

Discovered

Unseen

Stack (top at right): H | G
Visited: A, B, C, D, E, F, I, J



Step 18: Process node G

Step 18: Process node G

Legend

Complete

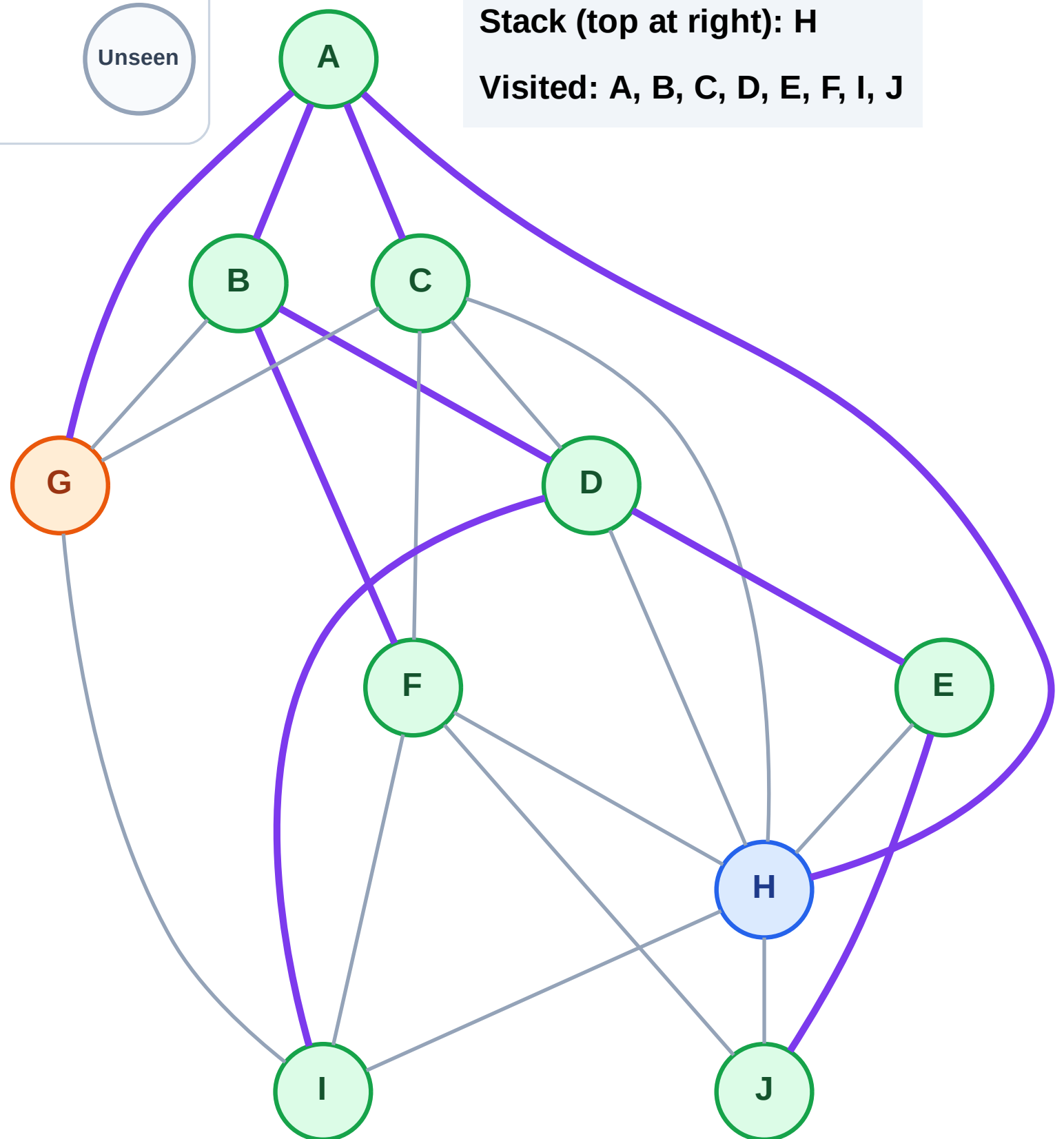
Processing

Discovered

Unseen

Stack (top at right): H

Visited: A, B, C, D, E, F, I, J



DFS Traversal

Step 19: No new neighbors from G

Legend

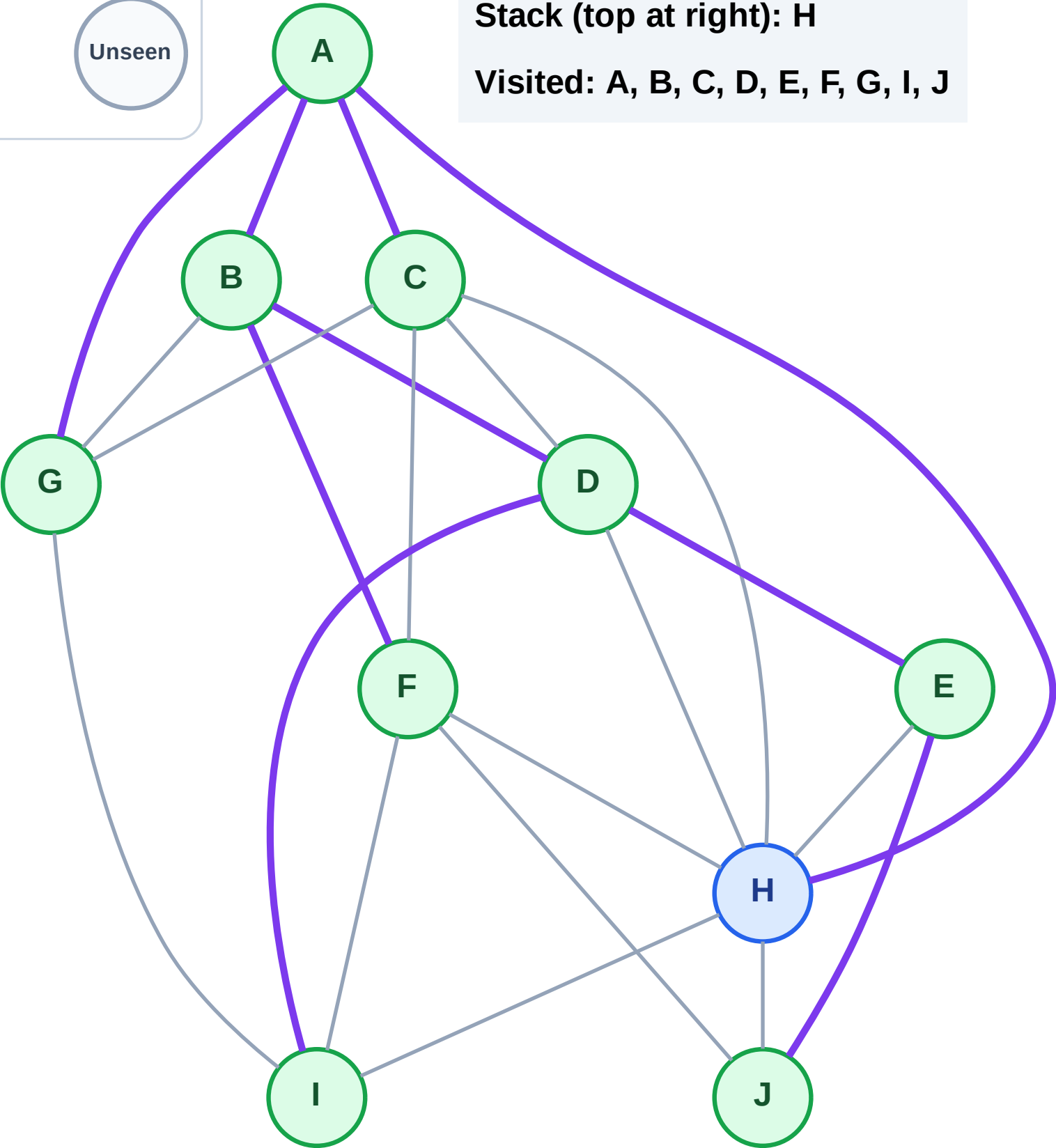
Complete

Processing

Discovered

Unseen

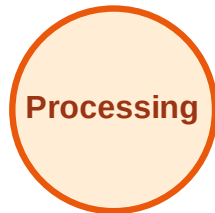
Stack (top at right): H
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

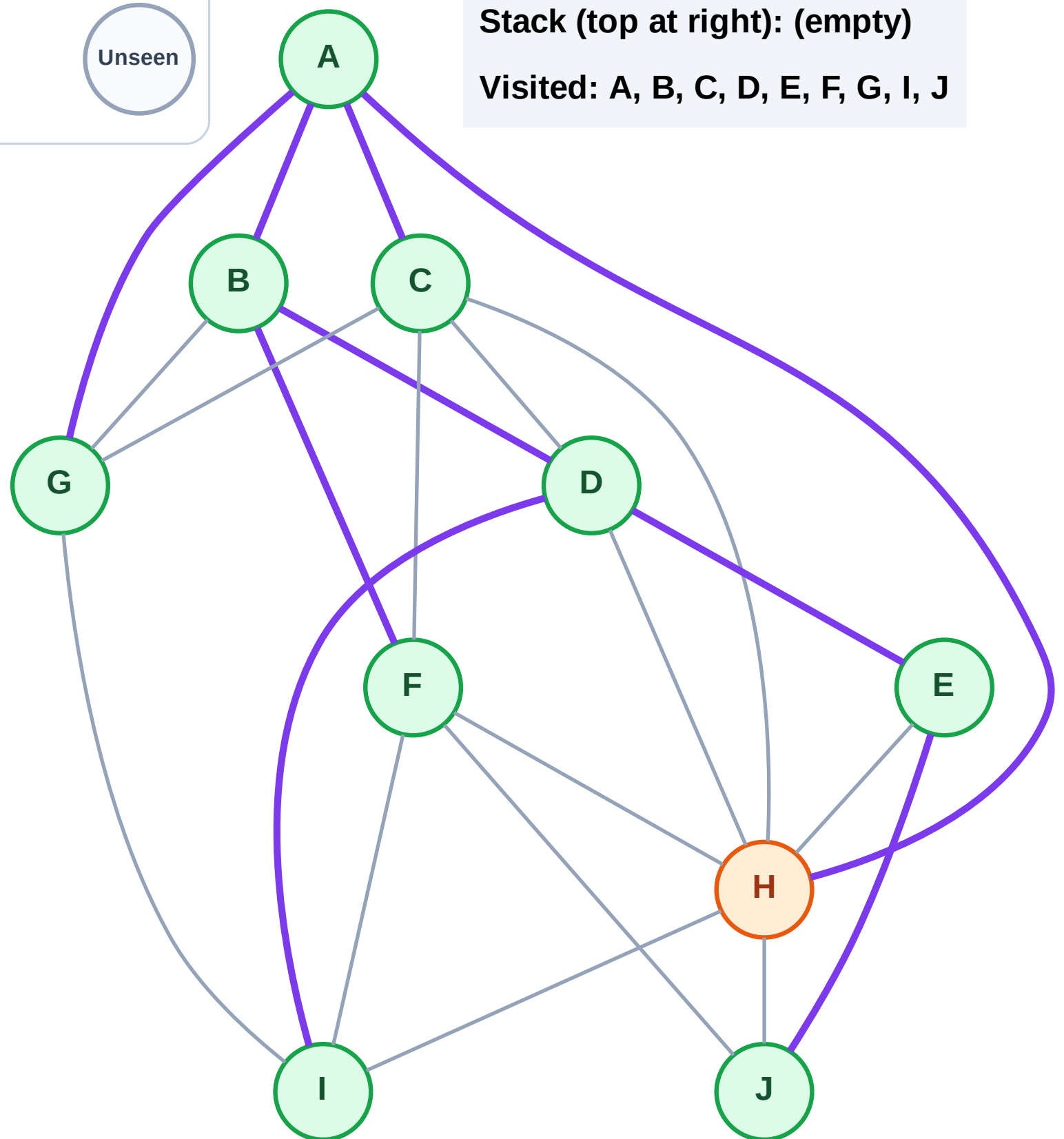
Step 20: Process node H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

